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Experimental Methods

This chapter describes how the measurements in this thesis were made reproducible. The central experimental object was not only the superconducting nanostructure itself, but the complete measurement chain around it: sample fabrication, cryogenic mounting, filtered electrical wiring, room-temperature instrumentation, digital acquisition, and numerical evaluation. Each part constrained the quality and interpretation of the measured I–V curves, dI–dV spectra, conductance maps, and extracted model parameters.

I first describe the fabrication of mechanically controlled break-junction samples. The process followed established group experience, but I adapted the design and fabrication protocol to improve sample stability, contacting, and reproducibility. This section therefore focuses on the steps that determined the mechanical and electrical properties of the final devices, rather than on a complete cleanroom manual.

I then describe the cryogenic and electrical setup used for transport measurements. The setup section follows the measurement signal from the sample environment to the room-temperature electronics. It discusses the dilution refrigerator, mechanical actuation, cold wiring, filtering, microwave coupling, voltage and current reconstruction, and grounding strategy. The emphasis is on the choices that affected electronic temperature, low-frequency noise, microwave irradiation, and the reliability of repeated spectra.

Finally, I describe the digital infrastructure that connected the physical setup to the data products analyzed in later chapters. This includes the Python-based instrument control, HDF5 data structure, evaluation pipeline, offset correction, and numerical model interfaces for tunnel-regime BCS fits and finite-transmission MAR analysis. Together, these sections define the experimental basis on which the results chapter builds.

more emphasis on multiple packages

keine citations in introduction, dafür motivation von vorgängern

1.1 Sample Fabrication

Single-electron transistors (SETs) consist of two junctions that define a small island capacitively coupled to a DC gate electrode. Typically, this geometry is realized using two oxide barriers in a wire fabricated by shadow evaporation [1, 2, 3].

In the present approach, one of these oxide barriers is replaced by a mechanically controlled break junction (MCBJ). Aluminum is chosen as the device material because it becomes superconducting at low temperatures [4, 5, 6]. In addition, an AC gate or strip line is integrated to ensure efficient and controlled coupling of microwave fields. Such a sample is shown in Figure 1.

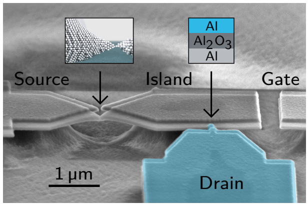

Figure 1 Scanning electron micrograph of the central sample region. The colored region highlights the top layer that forms one electrode of the oxide tunnel barrier. In addition, schematic insets illustrate the atomic contact at the MCBJ weak link and the cross-section of the tunnel barrier.

This section describes work carried out together with Patrick Raif as part of his project practical[1] [7].

The following subsections present the sample preparation process from complementary perspectives. First, the conceptual framework is introduced. Next, the main design choices and their rationale are discussed. The practical fabrication procedure itself, including practical considerations and typical challenges encountered in the laboratory, is given in the appendix in Subsection 1.4.

1.1.1 Concept

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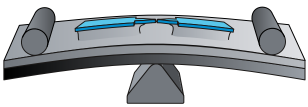


Figure 2 Schematic illustration of the principle of a MCBJ. Adapted from Ref [8].

In this subsection, I introduce the general concept of the sample fabrication process and its connection to the underlying device physics.

The sample design must satisfy several requirements simultaneously. Mechanically, it must provide a freestanding bridge on a bendable substrate so that an atomic contact can be formed in a controlled manner. Lithographically, it must enable suspended shadow-mask structures for double-angle evaporation. Electrically, it must define an island between two weak links and provide gate coupling. Finally, the design should allow efficient coupling of microwave fields.

The MCBJ is realized as a freestanding bridge-like structure on a bendable substrate, as illustrated schematically in Figure 2. The substrate can be bent

[1] A component of the master's curriculum completed within a semester.

either by pushing a stamp from below or by pressing counter rods from above. This bending elongates the bridge and allows the junction to be tuned into the atomic-contact regime. The mechanical setup of the MCBJ is described in more detail in Subsection 1.2.2.

A polyimide (PI) layer insulates the substrate from the structures produced in the subsequent steps. More importantly, the PI layer serves as a sacrificial layer. During later processing, it is partially etched, allowing the structure to become freestanding.

Next, a bilayer of positive electron-sensitive resist is applied. Electron-beam lithography (EBL) is then used to expose the desired pattern with high precision. After development, the exposed resist is removed, leaving behind the lithographically defined structure.

In addition to the MCBJ, the SET requires a second weak link, which is realized as an oxide barrier. A convenient method for fabricating this barrier is shadow evaporation [9, 10]. In this technique, two evaporation steps are carried out at different incident angles, producing a lateral offset between the deposited structures. Between the two depositions, a controlled oxidation step is performed without breaking vacuum. This yields overlapping regions with a metal–oxide–metal cross-section, as illustrated in Figure 3.

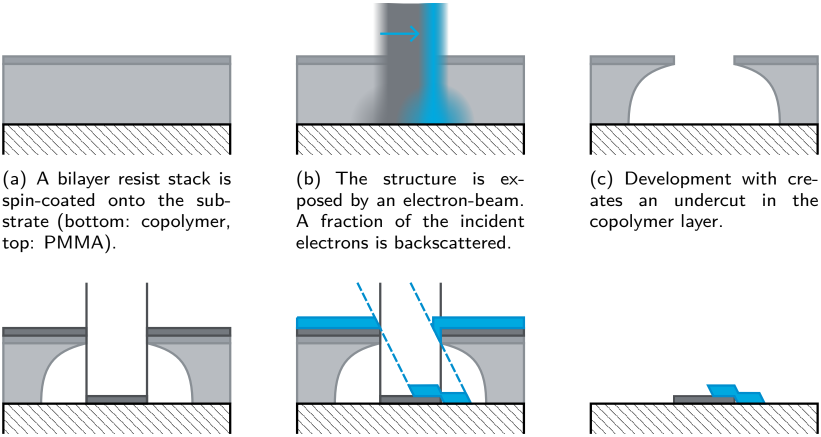
Since the aim is to fabricate a superconducting single-electron transistor (SSET), aluminum (Al) is used in both evaporation steps. This results in an Al break junction (BJ) and an Al–Al_2O_3–Al oxide barrier, as described by Thomas Lorenz [4] and Susanne Sprenger [5]. Alternatively, the second electrode material can be changed; for example, Al–Al_2O_3–Cu structures were successfully realized by Laura Sobral Rey [6].

In the final fabrication step, approximately 500 nm of the PI layer is removed by oxygen plasma etching. Because the etch proceeds essentially homogeneously, a pronounced undercut is created, as illustrated in Figure 1. This step releases the bridge and renders the break junction freestanding.

1.1.2 Design

This subsection discusses the fabrication process from the perspective of the underlying design decisions. While there is no single unique route to a working sample, individual choices can have important practical consequences. The following paragraphs summarize the reasoning behind the most relevant steps.

Bronze has long been used as a substrate material for MCBJ samples. A key reason is that it bends with comparatively little force while retaining excellent elastic behavior. This is crucial for MCBJs, where the junction configuration must be adjusted repeatedly and reproducibly. Other materials, such as copper, bend less uniformly and tend to deform mainly in the center beneath the stamp. This leads to larger angular changes and poorer elastic reproducibility. In addition, copper often shows non-monotonic bending behavior, which makes fine



(a) A bilayer resist stack is spin-coated onto the substrate (bottom: copolymer, top: PMMA).

(b) The structure is exposed by an electron-beam. A fraction of the incident electrons is backscattered.

(c) Development with creates an undercut in the copolymer layer.

(d) First metal deposition normal to the sample plane.

(e) Second deposition at an oblique angle. The second layer must be thicker to ensure electrical continuity.

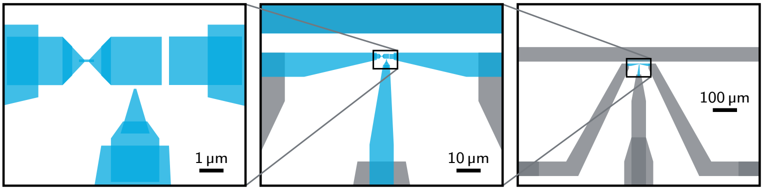
(f) The resist undercut facilitates lift-off after deposition.

Figure 3 Schematic cross-section of the fabrication process used to realize an oxide tunnel barrier by electron-beam lithography, and double-angle evaporation.

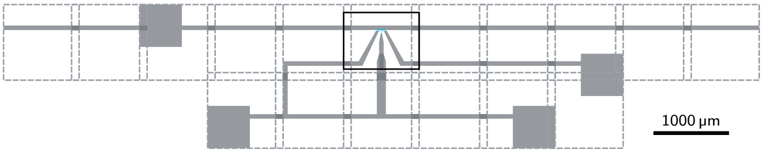
adjustment in the atomic-contact regime impractical. Kapton, by contrast, bends homogeneously along its length but does not allow sufficiently strong bending for the present MCBJ geometry. Copper–beryllium would be mechanically suitable, but its toxicity makes it undesirable in routine use. Finally, the substrate must be non-magnetic and contain as few magnetic impurities as possible. Considering all these requirements, bronze has proven to be the most suitable substrate material for this type of MCBJ over the years [8].

A polyamide (PA) layer is spin-coated onto the entire bronze wafer and converted into polyimide (PI) during a subsequent hard-bake step. Two resist layers are then applied. The thin top layer is optimized for high-resolution lithography and defines sharp feature edges. The thicker bottom layer serves two purposes: it lifts the top layer away from the substrate and promotes undercut formation during development. Two effects contribute to this undercut. First, electrons broaden as they pass through the resist stack, so the exposed volume widens with depth. In addition, backscattered electrons from the substrate can re-enter the resist from below and increase the effective exposure near the lower resist region. As a result, the bottom layer develops more rapidly and an undercut can form. Second, prolonged rinsing in isopropyl alcohol (IPA) after development in diluted methyl isobutyl ketone (MIBK) further enhances the undercut.

EBL is carried out using a Zeiss Crossbeam system that was modified for significantly faster beam blanking. The beam blanker operates synchronously with



(a) Central region, successively magnified, showing the break junction, the island, the oxide-barrier finger, the nearby DC gate, and the adjacent AC gate. Blue structures (■) are written in a 100 µm writing field, whereas gray structures (■) are written in 1000 µm writing fields.



(b) Overview of full pattern. Individual writing fields are indicated by dashed squares. Their overlap avoids stiching problems. The outer leads are arranged to simplify electrical contacting, and a shorting structure is included to protect the sample against electrostatic discharge.

Figure 4 Electron-beam lithography design of the sample at different length scales.

the beam deflection, allowing arbitrary structures to be exposed precisely within the selected field of view. For small high-resolution structures, a field size of 100 µm is typically used, whereas coarser and larger features are written in a 1000 µm field. Structures exceeding these dimensions are exposed over multiple adjacent fields by moving the sample stage. To ensure smooth transitions between neighboring fields, a deliberate overlap is included for stitching.

At the beginning of my doctoral work, only one of the two available EBL software packages, `Elphy Plus` by Raith, was functional and properly supported. The sample design therefore had to be reimplemented from scratch, since my predecessors had used a different software environment. The dimensions of the break junction and the gate follow the established geometry and are consistent with earlier implementations in Refs. [4, 5, 6]. In contrast, the finger defining the oxide barrier was modified to become wider toward its lower part. This reduces the freestanding length of the evaporation mask and thereby improves its mechanical stability. However, older resist material was found to collapse more easily during evaporation. In those cases, the finger does not reliably reach the island during the second evaporation step, which results in defective structures. The overall electron-beam lithography design is shown in Figure 4.

The larger leads and contact pads are arranged to simplify electrical contacting. In addition, an AC gate or strip line is placed at a distance of 8 µm to maximize microwave coupling while keeping the electron-beam writing time within reasonable limits. A shorting structure is also included to protect the sensitive tunnel barrier against electrostatic discharge. Finally, the central vertical lead is made wider to improve its visibility, which helps to align the break junction precisely with the stamp during mounting.

The evaporator used in this work has a comparatively long source-to-sample distance. As a result, the aluminum deposition is highly directional, although the effective deposition cross-section is reduced. The system offers two methods for aluminum evaporation: thermal evaporation and electron-beam evaporation. In thermal evaporation, a large current heats a small crucible until the aluminum evaporates. However, the crucible used for this method is fragile and must be loaded and run empty after each use to avoid crack formation. This procedure is laborious, prone to errors, and offers no clear advantage over electron-beam evaporation. In electron-beam evaporation, a focused electron beam locally heats the aluminum charge, resulting in a more efficient and reliable deposition process. For this reason, and in contrast to Refs. [4, 5, 6], electron-beam evaporation was used throughout this work.

The oxidation step between the two evaporations is performed in the load lock of the evaporator at an oxygen pressure of approximately 3.0(2) mbar for 3 min [6]. In the past, this step was found to be insufficiently reproducible. It is therefore monitored carefully in the present process [5, 11].

Reactive ion etching (RIE) is carried out using the *PlasmaPro 80 ICP RIE system* from Oxford Instruments. This system was newly introduced when I began my doctoral work, so the fabrication of break junctions with it had to be established from scratch, including the development of a suitable etching recipe. Although this tool is not primarily designed for highly homogeneous etching, acceptable uniformity can be achieved by combining low table radio-frequency (RF) power, high inductively coupled plasma (ICP) RF power, and high chamber pressure. The ICP source generates a high-density plasma by coupling energy into the chamber via an electromagnetic field, whereas the RF power applied to the table controls the ion energy at the sample surface. A heated sample table further improves etch homogeneity. The etched depth is monitored in real time by laser interferometry, which allows precise process control.

The final step is electrical contacting of the sample. Wedge bonding is unsuitable because the bonding wire can press through the pad and penetrate the soft PI layer, thereby causing a short. Instead, thin copper wires are attached to the pads using conductive silver paste. This approach provides reliable electrical contact without damaging the underlying structure.

1.2 Experimental Setup

The experiments in this thesis require a setup that can form nanometer-scale superconducting junctions, tune them in situ, and measure nonlinear transport at energies of only a few tens of microelectronvolts. The relevant object is therefore not only the lithographic device. It is the complete measurement environment around it. The cryostat determines temperature, cooling power, vibration level, and thermal anchors for the wiring. The mechanically controlled break junction (MCBJ) opens a metallic constriction from a closed contact into the few-channel and tunnel regimes. The cold electrical environment defines the bias circuit, filters external radiation, thermalizes the electronic degrees of freedom, and sets how microwave power reaches the sample. The instrumentation and grounding determine how voltages are generated, amplified, digitized, synchronized, and isolated from unwanted ground paths.

These parts cannot be treated as independent accessories. A stable atomic contact is only useful if the cryostat vibration level is low enough to hold it during a slow sweep. A low base temperature is only useful if the electrical lines are filtered and thermalized well enough that the electronic temperature follows the cold stage. A nominal bias voltage is meaningful only if the voltage division between reference resistors, filters, leads, and the sample is measured rather than assumed. A microwave source power is likewise not the same as a calibrated local microwave voltage at the junction. The setup therefore defines both the experimental control parameters and the limits of their interpretation.

The platform was inherited from earlier MCBJ experiments in the group, but it was modified substantially during this work for superconducting devices, microwave irradiation, local thermometry, and more robust repeated cooldowns. Mechanical parts were replaced, gearbox ratios were redistributed, fragile wiring was removed, filter and reference-resistor layouts were revised, and the grounding configuration was selected empirically from noise measurements. These practical details determine whether the data are reproducible. They also state which assumptions are justified when later chapters interpret I–V curves, dI–dV spectra, conductance maps, and microwave-induced features.

I organize this section by experimental constraint rather than by component list. First, I describe the cryogenic environment, including precooling, condensation, dilution refrigeration, vibration, and heat load. Second, I describe the MCBJ mechanics and the conversion from motor motion to the relative contact-control coordinate. Third, I define the cold electrical measurement environment, including voltage and current reconstruction, filtering, and microwave coupling. Finally, I describe the instrumentation and grounding, including the practical cryostat ground reference and the isolation choices used for low-noise measurements. Together, these subsections record the physical setup and the measurement model used throughout the thesis.

1.2.1 Cryogenic Environment

The measurements were carried out in an LD400 Bluefors dry dilution refrigerator. The cryostat has to solve three coupled problems before a superconducting nanostructure can be measured reproducibly. It must remove heat from room temperature to the millikelvin regime, thermalize the wiring and sample environment, and keep mechanical motion small enough that an atomic contact remains stable during slow conductance tuning. The relevant cooling chain is therefore described here in operational order. The pulse tube provides closed-cycle precooling and defines much of the vibration environment. The Joule–Thomson stage condenses the circulating helium mixture after sufficient precooling. The dilution unit then supplies the continuous millikelvin cooling power at the mixing chamber.

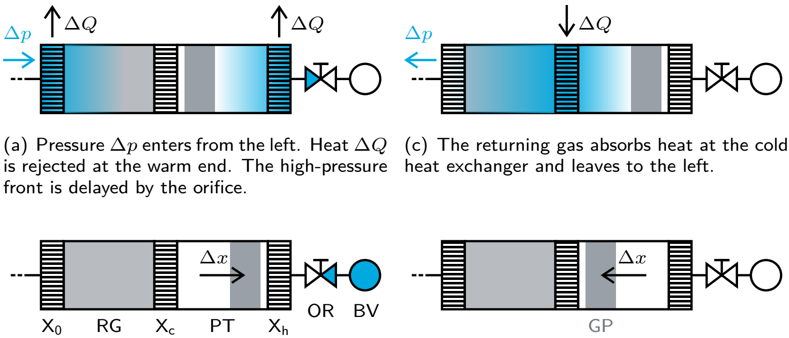
Pulse Tube Cryocooler

The first active cooling stage is a two-stage Gifford–McMahon-type (GM-type) pulse tube. It is a closed-cycle helium refrigerator. A room-temperature compressor and rotary valve generate a pressure oscillation, while the cold head contains no mechanical displacer. This distinction is important for the MCBJ experiment. The pulse tube removes the dominant heat load, cools the radiation shields, provides thermal anchors for wiring, and brings the helium mixture to the few-kelvin range. At the same time, the pressure oscillation and rotary valve remain a source of mechanical vibration.

Figure 5 shows the basic operation of a single-stage orifice pulse tube, following Refs. [13, 12]. The compressor drives helium back and forth through the aftercooler, regenerator, cold heat exchanger, pulse-tube volume, warm heat exchanger, and the flow impedance to the buffer volume. The gas does not circulate through the cold head in one direction. It oscillates. The regenerator stores heat during one part of the cycle and returns it during another, which maintains the temperature gradient between the warm and cold ends.

Cooling is produced by the phase relation between pressure and gas displacement. In a conventional GM cooler, this phase relation is imposed by a cold mechanical displacer. In a pulse tube, the gas column itself acts as a virtual piston. The warm-end impedance and buffer volume delay the flow relative to the pressure oscillation. With the correct timing, gas rejects heat at the warm heat exchanger and absorbs heat at the cold heat exchanger. The useful cooling can therefore be understood as an enthalpy flow through an oscillating gas column.

The LD400 uses two cascaded pulse-tube stages, as sketched in Fig. 6. The first stage removes heat at intermediate temperature and provides a thermal anchor near 40 K. The second stage is precooled by the first and reaches the few-kelvin range. These stages are not only refrigeration points. They are also the first thermal anchors for radiation shields, cabling, and the incoming mixture



(a) Pressure Δp enters from the left. Heat ΔQ is rejected at the warm end. The high-pressure front is delayed by the orifice.

(c) The returning gas absorbs heat at the cold heat exchanger and leaves to the left.

(b) The virtual gas piston is displaced, while the buffer volume delays the gas flow.

(d) The virtual gas piston moves back toward equilibrium before the next cycle begins.

Figure 5 Schematic of pulse-tube cooling. The stage consists of an aftercooler (X_0), regenerator (RG), cold heat exchanger (X_c), pulse tube (PT), warm heat exchanger (X_h), virtual gas piston (GP), orifice (OR), and buffer volume (BV). High-pressure regions are indicated in blue (■). Adapted from Ref. [12].

line. Their performance therefore affects both the cooldown time and the heat load that must later be handled by the dilution unit.

Mechanical decoupling is part of the cryostat design because pulse-tube-driven motion can couple to the break junction. The cryostat frame is supported by pneumatic air springs at the four corners, operated at about 6 bar and the suspended assembly has a large mass. The mass and air springs reduce the transmission of building and compressor vibrations to the MCBJ. They do not remove all motion, but they make slow contact tuning practical. In this configuration, contacts could remain stable on the scale of weeks as long as the cryostat was not mechanically disturbed. A small fraction of contacts still showed atomic rearrangements during individual measurements. I treat this as an intrinsic limitation of atomic-scale junctions, not only as a cryostat-stability issue.

The pulse-tube system is driven by an oil-lubricated room-temperature helium compressor.[2] Low-pressure helium returning from the cold head is compressed, cooled, purified, and supplied again as high-pressure helium. Oil improves sealing and removes heat from the compressor, but it must not reach the cold head. Oil separators, coalescing filters, and adsorbers therefore remove oil droplets, mist, and residual contaminants before the helium returns to the cryostat. This purification chain is essential because impurities can freeze out at cryogenic temperatures and degrade the regenerator or block small flow impedances [14].

[2] Cryomech, `CPA1110W2FNCE-21770300001`.

Compressor maintenance became relevant during this work. The adsorber was replaced, and the compressor heat exchanger required repeated repairs due to leaks. One practical cause was the cooling-water supply. The compressor was cooled with water from Lake Constance, which is very soft and can corrode copper cooling-water circuits if it is not chemically conditioned. Dirt particles and shell debris in insufficiently filtered lake water can additionally abrade the piping [15]. Eventually, a leak between the cooling-water circuit and the helium circuit led to replacement of the compressor package. This history is included because compressor degradation or helium contamination can reduce pulse-tube performance before a problem is visible at the sample stage.

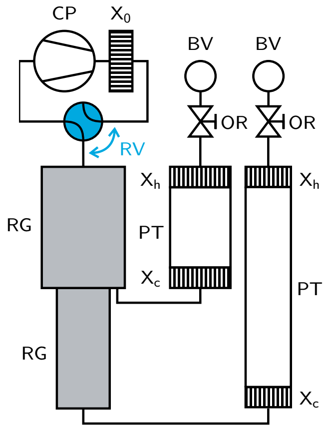


Figure 6 Schematic of a two-stage Gifford–McMahon-type pulse tube. The rotary valve (RV) alternately connects the cold head to the high- and low-pressure sides of the compressor (CP). The first stage provides an intermediate thermal anchor, while the second stage is precooled by the first stage. Adapted from Ref. [13].

Joule–Thomson Cooling

After pulse-tube precooling, the circulating ^{3}He / ^{4}He mixture must be condensed before it can feed the dilution unit. The mixture cannot simply be expanded from room temperature into the cold part of the cryostat, because the incoming gas would carry too much enthalpy. It is first thermalized at the pulse-tube stages and by lower-temperature return flows. It then passes through a flow impedance. This is the Joule–Thomson stage of the refrigerator [13, 16].

Joule–Thomson cooling is the temperature change of a real gas during throttling at approximately constant enthalpy H,

$$H = U + pV\,, \qquad \Delta H \approx 0\,, \tag{1}$$

with internal energy U, pressure p, and volume V. During throttling through a narrow restriction, no useful mechanical work is extracted and ideally no heat is exchanged within the impedance. The gas therefore changes pressure without changing its enthalpy appreciably.

For a small pressure drop, the corresponding temperature change can be written as

$$\Delta T \approx \mu_{\mathrm{JT}}\,\Delta p\,, \tag{2}$$

where μ_{JT} is the Joule–Thomson coefficient, which is given by

$$\mu_{\mathrm{JT}} = \left(\frac{\partial T}{\partial p}\right)_H = \frac{V}{C_\mathrm{p}}(\alpha T - 1)\,. \tag{3}$$

Here C_p is the heat capacity at constant pressure and α is the thermal expansion coefficient. The Joule–Thomson coefficient is the slope of an isenthalpic path in the temperature-pressure plane. Its sign determines whether a pressure drop cools or heats the gas. For an ideal gas, enthalpy depends only on temperature, so an isenthalpic expansion would not change the temperature. Joule–Thomson cooling is therefore a real-gas effect.

The sign of μ_JT changes at the inversion temperature, as shown in Fig. 7 for ^{4}He. Helium has an unusually low inversion temperature, so room-temperature helium would not be cooled by Joule–Thomson expansion. The exact values differ for the circulating ^{3}He / ^{4}He mixture, but the practical requirement is the same. The gas must be precooled before throttling can be used for condensation. The pulse tube is therefore not just a convenience stage. It brings the mixture into the thermodynamic regime where expansion removes heat instead of adding it.

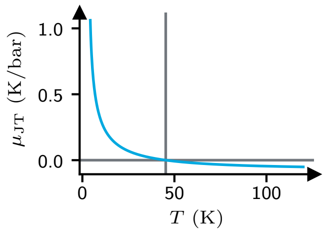
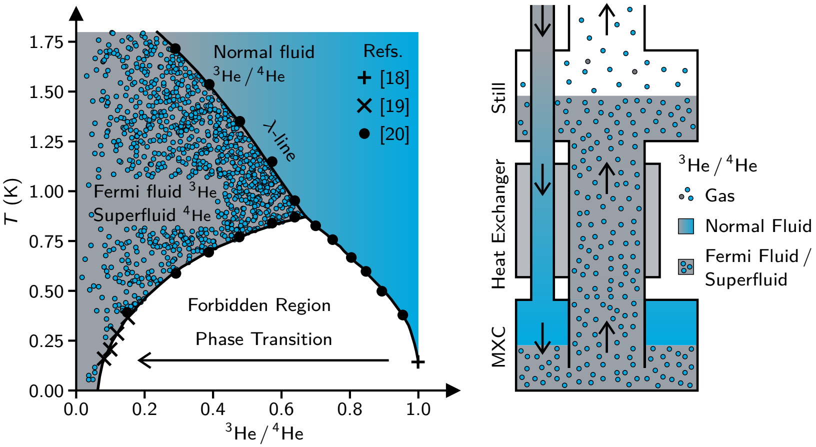


Figure 7 Joule–Thomson coefficient of ^{4}He as a function of temperature. The curve was calculated for a pressure of 1 bar using CoolProp [17] and shows the change of sign at the inversion temperature.

In the dry dilution refrigerator, the incoming high-pressure mixture is cooled stepwise before it reaches the Joule–Thomson impedance. It is anchored at the pulse-tube stages and exchanges heat with colder return flows in the lower-temperature part of the dilution unit. After this precooling, the pressure drop across the impedance lowers the temperature further and condenses the mixture. The Joule–Thomson stage thus connects pulse-tube precooling to dilution refrigeration by converting a precooled, pressurized gas flow into a cold liquid mixture for continuous circulation.

Initial condensation and regular circulation have different failure modes. During condensation, the mixture line is operated at overpressure to drive gas through the impedance and condense it. Leaks can therefore lose mixture to the vacuum space or laboratory environment. During regular circulation, by contrast, the condensing line is usually below atmospheric pressure. Small leaks are then more likely to draw residual gas or contamination into the circulation path. This is dangerous because the Joule–Thomson impedance is narrow and can become clogged by gases or moisture that freeze out at cryogenic temperatures. For this reason, the mixture handling line is passed through a liquid-nitrogen trap before condensation or circulation.

Once the mixture has condensed and circulation is established, the same impedance remains part of the continuous flow path. The pressure drop and counterflow heat exchangers supply cold liquid mixture to the lower-temperature stages, while the actual millikelvin cooling is produced in the mixing chamber.



(a) Phase diagram of liquid ^{3}He / ^{4}He mixtures. The white region is forbidden for a homogeneous liquid, because mixtures in this range lower their free energy by separating into a ^{3}He-rich concentrated phase and a ^{4}He-rich dilute phase. The λ-line marks the superfluid transition of the ^{4}He-rich phase. Adapted from Ref. [21].

(b) Dilution-unit circulation. The incoming ^{3}He-rich stream is precooled, enters the mixing chamber (MXC), and cools it as ^{3}He crosses into the dilute phase. The still evaporates ^{3}He and closes the cycle.

Figure 8 Dilution refrigeration with a ^{3}He / ^{4}He mixture. (a) The phase diagram shows the miscibility gap and the finite low-temperature ^{3}He solubility of the dilute phase. (b) The circulation schematic shows how continuous removal of ^{3}He from the dilute phase keeps the mixing chamber cold.

Dilution Refrigerator

The dilution unit is governed by the low-temperature phase diagram of the helium mixture rather than by a further gas-expansion step. Figure 8(a) shows that a homogeneous liquid is stable only outside the phase-separation boundary. The white “Forbidden Region” is therefore not an impossible total mixture composition. It is a miscibility gap. If the average ^{3}He concentration and temperature place the mixture inside this region, the homogeneous liquid has a higher free energy than two coexisting liquids. The mixture therefore separates into a ^{3}He-rich concentrated phase and a ^{4}He-rich dilute phase with compositions on the phase boundary at the same temperature [21, 16].

This phase separation is useful because the dilute phase does not become pure ^{4}He as the temperature approaches zero. Instead, it retains a finite ^{3}He concentration of about 0.066, while the concentrated phase approaches nearly pure ^{3}He. The mixing chamber exploits exactly this property. Its phase boundary separates the upper concentrated phase from the lower dilute phase. When ^{3}He atoms cross from the concentrated phase into the dilute phase, they enter a state with larger enthalpy. The required energy is taken from the mixing chamber, so the transfer cools the sample stage. The refrigerator can keep cooling only if this transfer is driven continuously. Without circulation, the phases would approach equilibrium and the cooling power would vanish.

The circulation path in Fig. 8(b) shows how this non-equilibrium condition is maintained. The incoming ^{3}He-rich stream is precooled by the outgoing cold dilute stream in the counterflow heat exchangers and enters the mixing chamber on the concentrated side of the phase boundary. After crossing into the dilute phase, the ^{3}He is carried upward with the dilute mixture toward the still. At the still temperature, ^{3}He has a much higher vapor pressure than ^{4}He, so still heating evaporates ^{3}He preferentially. The pumps remove this ^{3}He-rich gas, the room-temperature gas-handling system cleans it, and the condensing line returns it to the cryostat. The cooling power is therefore set mainly by the ^{3}He circulation rate and by the heat-exchanger efficiency. The heat exchangers determine how much of the outgoing cooling is recovered before the incoming stream reaches the mixing chamber [13, 16].

Typical preparation and cooldown values are useful diagnostics for later operation. Before cooldown, the cryostat vacuum space was pumped for up to 24 h and reached pressures of about 6×10^{-3} mbar. During cryogenic operation, the insulating vacuum reached about 2×10^{-6} mbar. A full cooldown to the few-kelvin regime, including the magnet, took about 36 h. Under typical operating conditions the nominal 40 K stage stabilized near 47 K, while the 4 K stage reached about 3.2 K. Once the system was at 4 K, condensation of the mixture took about 1 h, and the subsequent cooldown from 4 K to base temperature took about 3 h.

The LD400 system is specified with a guaranteed base temperature of 10 mK and an expected base temperature of 8 mK. The specified cooling powers at the mixing-chamber flange are 14 μW at 20 mK, 400 μW at 100 mK, and 575 μW at 120 mK [22]. These values are manufacturer specifications for the cryostat and were not remeasured for the installed experiment. With the installed MCBJ experiment, wiring, filters, thermometers, and microwave infrastructure, the sample and mixing-chamber stages could reach about 30 mK, while the still was operated at about 1 K. A normal automated warm-up with the Bluefors warm-up script took about 2.5–3 days.

I did not optimize the ^{3}He circulation rate for the absolute lowest mixing-chamber temperature during this work. The limiting factor for the transport measurements was not the final few millikelvin of lattice temperature. The setup could be operated comfortably between about 30 mK and 40 mK, and the MCBJ could be moved step by step without leaving this temperature regime. The largest temperature excursions occurred during microwave irradiation. Depending on the applied microwave power and coupling path, the sample-stage temperature could rise to several hundred millikelvin. In extreme cases it approached 1 K. Improving the last part of the base-temperature performance was therefore less relevant than controlling microwave heating and maintaining a well-thermalized, well-filtered electrical environment. These conditions were more important for obtaining sharp spectra. The measured sample-stage temperature should therefore not be identified automatically with the electronic temperature relevant for the junction, which can be higher because of residual radiation, imperfect line thermalization, bias dissipation, or microwave-induced heating. Earlier optimization of the same cryostat and MCBJ platform by Martin Prestel reached temperatures below 30 mK before the modifications described here [23].

Together, these stages define the thermal environment of the experiment. The pulse tube provides the dominant cooling power at intermediate temperatures, the Joule–Thomson stage condenses the helium mixture and enables continuous circulation, and the dilution unit provides the millikelvin cooling power at the mixing chamber. The sample, filters, thermometers, and wiring are thermally anchored to the corresponding cryostat stages. The cryostat therefore does not only set the lattice temperature of the device. It also determines how well electrical lines are thermalized, how much external radiation reaches the contact, and how much mechanical motion is transmitted to the break junction. These limits directly affect the sharpness of the spectra, and the reproducibility of contact tuning.

1.2.2 Mechanically Controlled Break Junction

The mechanically controllable break-junction (MCBJ) setup provides the in situ control coordinate of the experiment. Its task is to convert a macroscopic room-temperature motor motion into a reproducible, much smaller elongation of the metallic constriction at millikelvin temperature. The sample is mounted in a three-point bending geometry, as shown in Fig. 9. Two fixed counter supports hold the substrate, while a central pushing element bends it from the opposite side. In this implementation the counter supports remain fixed and the central element is displaced. The suspended metallic bridge on the elastic substrate then elongates by only a small fraction of the macroscopic bending motion. This mechanical reduction allows atomic-scale contacts to be formed, opened, and held over long measurement times.

The cryogenic mechanics used here are based on the LD400-compatible design developed by A. Fischer and M. Prestel [23] and built by the scientific workshop.[3] The rotary motion from the drive train is converted into linear motion by a differential screw. The screw combines an M4 and an M5 thread, so one full revolution advances the bending mechanism only by the difference between the two thread pitches. The resulting macroscopic feed is 0.1 mm per revolution. This differential screw is therefore the final mechanical interface between the cryogenic rotary drive train and the three-point bending mechanism.

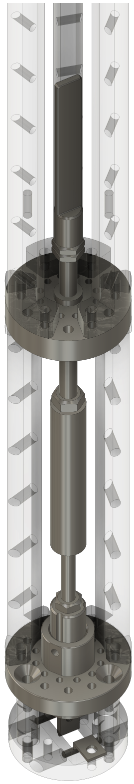


Figure 9 Rendering of the differential screw, central pushing element, counter supports, and sample holder used for the cryogenic MCBJ mechanics.

I did not independently calibrate the conversion from motor motion to junction elongation. The absolute elongation of the atomic contact is therefore not used as a calibrated variable in this thesis. Instead, the contact position is treated operationally. The motor encoder reports large integer values up to about 1.8×10^9, which are not useful for describing contact tuning. The laboratory control software maps the motion to a relative coordinate from 0 to 18. This range corresponds to 30 revolutions of the differential screw and therefore to 3 mm of macroscopic screw feed. It should be read as a reproducible opening and closing coordinate, while the conductance response provides the physical calibration of the contact state.

[3] Scientific workshop, 07062016.

The cryogenic drive train has to transmit controlled rotary motion from room temperature to the sample space without creating excessive heat load or mechanical stress. It is therefore segmented and thermalized at several cryostat stages. Hollow stainless-steel tubes connect the individual gear stages.[4] The gearboxes transmit motion and interrupt the thermal path along the shaft.

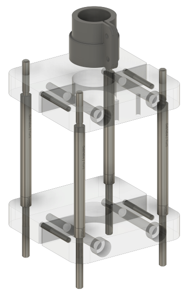


Figure 10 Rendering of the redesigned room-temperature mounting plates for the motor and vacuum feedthrough.

The inherited drive train used four commercial gearboxes with equal reduction. During this work, repeated gearbox failures showed that this distribution loaded the upper cold gearboxes too strongly. All cryogenic gearboxes are Faulhaber Series 22/2K gearboxes. I replaced the gearhead close to the motor by a 1:9.7 reduction and the first two cryogenic gearboxes by 1:5.4 reductions. The remaining 1:3.1 gear stages were left unchanged. This redistribution places less torque on the first cryogenic gearbox. If a lower stage blocks or becomes stiff, the upper gearbox is then less likely to deform the thin drive rod or decouple from the following shaft. The additional rotations at the 40 K and 4 K stages are acceptable, because these stages have much larger cooling power than the millikelvin stages.

Several room-temperature drive components were redesigned in parallel with the cold gearbox changes. In the previous motor mount, the clamping element was a massive plastic disk with a central bore and a narrow radial slit. Closing the slit mainly deformed the plastic body instead of reducing the bore diameter. The thread was also cut directly into the plastic and was damaged by the force needed for clamping. As a result, the mount did not hold the motor or the vacuum feedthrough reproducibly. I replaced it with the two-piece clamp[5], designed by O. Bruttel and myself, shown in Fig. 10. Separate clamps now hold the motor and vacuum feedthrough, and rods connect the two plates. This gives a stiffer and more reproducible alignment of the room-temperature drive. The motor and gearhead are also held by a 3D-printed clamp instead of being glued in place.

The vacuum feedthrough and slip clutch were treated as part of the same reliability problem. The rotary feedthrough[6] was lubricated regularly with vacuum grease[7] to keep friction low and to avoid stick–slip motion or excessive motor load. In the original slip clutch, the clamping threads were cut into plastic. Repeated compression deformed these threads and shifted the clutch away from the rotation axis. The resulting misalignment increased the motor load and led to premature motor failure. The plastic threads were replaced by brass inserts,

[4] Faulhaber, `401.4659.04 "Ritzel" 2.0x4.35x4.7 Z12 Weld`, welded to stainless-steel tubes with 4 mm outer diameter and 0.1 mm wall thickness.
[5] Scientific workshop, 13042026.
[6] Pfeiffer Vacuum, `DD 016 A`, rotary feedthrough, `DN 16 ISO-KF`.
[7] Fomblin, 1234567890.

which improved centering and made the coupling stable under repeated clamping.

Assembly procedure was as important as the replacement parts. The rods inside the cryostat were fixed while rotating slowly, so small angular or lateral offsets could relax before the train was locked in place. This reduces the risk that residual misalignment freezes into the completed drive train and blocks motion at low temperature. Before cooldown, each gearbox was stripped of its original lubricant, cleaned overnight in an ultrasonic acetone bath, rinsed with isopropanol, and immediately before cooldown heated with a hot-air gun. This removes residual solvent and adsorbed moisture. The preparation is necessary because lubricant, solvent, or water residues can freeze and block the gears at cryogenic temperature.

Intentional motion at low temperature provides a useful diagnostic of the mechanical chain. If the drive train is connected and the cold mechanism moves, stepwise actuation typically produces a visible increase in sample-stage temperature. In some tests, the sample stage heated up to about 1 K. This heating is not desirable during sensitive measurements, but it shows that mechanical work reaches the cold break mechanism. During such tests the gas-handling system has to be monitored, because the added heat load on the still can increase the return pressure and may cause overpressure if the motion is too aggressive or repeated too quickly.

The final configuration was reached through iterative commissioning rather than a single repair. Each mechanical change had to be tested in a full cryogenic cycle. A failed test usually meant another warm-up, repair or replacement, pump-down, and cooldown. This made the mechanical optimization one of the slower parts of the experiment, but it also made the final operating procedure reproducible. After the gearbox distribution, motor mount, slip clutch, feedthrough lubrication, and assembly procedure had been revised, the mechanics could be operated reliably during cooldown and at base temperature. In the final cooldown, I could break and tune the sample without further mechanical problems. The practical takeaway is that the MCBJ position is a reproducible conductance-control coordinate, while the absolute atomic displacement remains inferred from the measured transport response.

1.2.3 Electrical Measurement Environment

The electrical environment defines how the device is biased and how the recorded voltages are converted into physical observables. It is not a passive connection between the sample and the instruments. The wiring sets the source impedance, the noise environment, the thermalization of the electronic degrees of freedom, and the microwave coupling to the junction. These constraints are especially important for an MCBJ experiment, because the device resistance changes by orders of magnitude while the same cryogenic wiring remains in place.

In this section, I describe the electrical setup from the point of view of the later data analysis. I first define how the sample voltage and current are reconstructed from measured voltage drops. I then describe the cold wiring and filters that make this reconstruction meaningful at millikelvin temperature. Finally, I discuss the microwave lines and thermometry. The room-temperature instruments and grounding strategy are treated separately in Sec. 1.2.4. The goal is to state which quantities are calibrated, which quantities are measured only effectively, and which assumptions enter the interpretation of *I*–*V* curves, *dI*–*dV* spectra, and microwave-driven maps.

Measurement Schematic

The transport circuit is operated in a quasi-dc mode. A programmed voltage V_{bias} is applied to the bias line, but this source voltage is not treated as the sample voltage. The cold reference resistors, filters, leads, contacts, and the device itself form a bias divider. The relevant observables are therefore reconstructed from measured voltage drops. The sample voltage V_{sample} is measured across the device region, and the current is inferred from the voltage drop V_{ref} across a known reference resistor.

In the standard configuration, two cold reference resistors are placed symmetrically on the two sides of the sample. This keeps the sample potential close to the cryostat ground during a bias sweep and reduces common-mode offsets in the voltage readout. The voltage V_{ref} is measured across one of these resistors. Its measured base-temperature resistance is

$$R_{\text{ref}} = 101.473\,\text{k}\Omega\,. \tag{4}$$

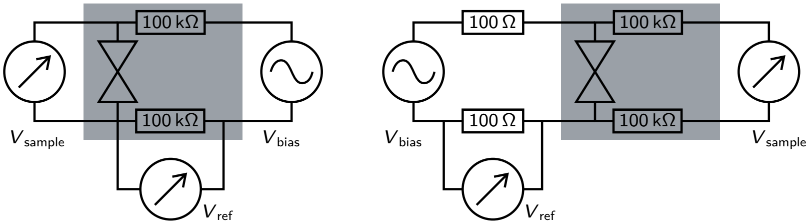
The second cold reference resistor contributes to the total series resistance of the bias path, but it is not part of the measured reference-voltage drop.

A useful measure of the instantaneous bias condition is

$$\eta = R_{\text{series}}\,/\,R_{\text{sample}}\,, \tag{5}$$

where R_{sample} is the instantaneous device resistance and R_{series} is the remaining resistance in the bias path,

$$R_{\text{series}} \simeq 2R_{\text{ref}} + R_{\text{filters}} + R_{\text{leads}} + \dots\,. \tag{6}$$



(a) Schematic for high-impedance samples.

(b) Schematic for low-impedance samples with an additional warm reference resistor.

Figure 11 Measurement schematics used to reconstruct sample voltage and current. A quasi-dc bias voltage V_{bias} is applied to the bias line, while V_{sample} and V_{ref} are measured simultaneously. The opposing triangles indicate the sample, and the gray region (■) marks the cold part of the circuit. (a) In the high-impedance configuration, two cold reference resistors are placed symmetrically around the sample. (b) In the low-impedance configuration, the voltage-sense and bias wiring are exchanged so that a larger fraction of the programmed bias drops across the sample. An additional warm reference resistor provides the current readout.

In the limit $\eta \ll 1$, most of the applied voltage drops across the sample and the circuit approaches voltage bias. In the opposite limit, $\eta \gg 1$, the series resistance dominates and the circuit approaches current bias. The MCBJ moves between these limits during operation. A closed junction can have a resistance of order 100 Ω, a few-channel atomic contact has a resistance of order 12.9 kΩ, and the tunnel regime starts roughly around 100 kΩ. Since the standard filtered bias environment already contains about 200 kΩ from the two cold reference resistors, the same wiring can represent a voltage-biased, mixed, or more current-biased environment depending on the junction state.

This changing bias condition is the reason why both V_{sample} and I_{sample} are reconstructed from measured voltages rather than from the programmed source value. Since V_{ref} is measured across one cold reference resistor, the current is

$$I_{\mathrm{sample}} = V_{\mathrm{ref}} \,/\, R_{\mathrm{ref}} \,. \tag{7}$$

In a linearized resistive approximation, the voltage and current would follow

$$V_{\mathrm{sample}} \simeq V_{\mathrm{bias}}\, R_{\mathrm{sample}} \,/\, R_{\mathrm{total}} \,, \tag{8}$$

$$I_{\mathrm{sample}} \simeq V_{\mathrm{bias}} \,/\, R_{\mathrm{total}} \,. \tag{9}$$

These expressions are useful for intuition, but the actual analysis uses the simultaneously measured voltage drops. This is essential for nonlinear superconducting transport, where the differential resistance changes with bias and small deviations from an ideal bias condition can distort the apparent *I–V* curve.

Figure 11(a) shows the standard configuration used for high-impedance tunnel and few-channel measurements. For very low sample impedances, the cold series resistance limits the voltage that reaches the sample. A substantial part of the programmed bias then drops across the reference resistors instead of the device. The accessible sample-voltage range can become too small to drive the junction out of the zero-voltage supercurrent branch or to reach spectroscopic features such as the superconducting gap.

In this regime, I used the modified configuration shown in Fig. 11(b). The voltage-sense and bias wiring are exchanged, and the symmetric cold resistors in the bias line are reduced to nominally 100 Ω each. The current is then read out using an additional warm reference resistor. This increases the accessible sample-voltage range, but it moves part of the current readout to room temperature and adds Johnson–Nyquist noise from the warm environment [24, 25]. I therefore used the low-impedance configuration only when the larger voltage range was more important than the added noise.

The measurement schematic defines the electrical meaning of each recorded trace. The programmed source voltage is only a control parameter. The physical observables used later are the reconstructed sample voltage, the reconstructed current, and the bias condition set by the ratio of sample resistance to series resistance.

Wiring inside Cryostat

The cryostat wiring has to satisfy two requirements at the same time. It must define a low-noise electrical environment for high-impedance tunnel devices and few-channel atomic contacts, and it must not carry excessive heat from room temperature to the millikelvin stage. The first requirement motivates symmetric wiring, cold reference resistors, shielding, and filtering of external high-frequency noise. The second requirement motivates low-thermal-conductance cables and careful thermal anchoring. The wiring concept follows earlier MCBJ and single-electron-transistor setups in the group [4, 23], but several parts were changed for this experiment.

Figure 12 shows the custom wiring used near the sample. The low-frequency wiring contains six functional line groups: sample-voltage readout, bias, current readout, DC gate, thermometer, and heater. The DC gate uses two physical lines that are shorted at the lowest practical point. This gives diagnostic and backup access while the sample sees one gate terminal. From room temperature to the base-temperature region, these lines use the commercial Bluefors cryostat cabling, in contrast to earlier implementations [23]. At the base-temperature stage, the lines are continued with double-twisted coaxial cable[8] with brass inner

[8] GVL Cryoengineering, `GVLZ 169`, connected with two-pole LEMO, `FGG.00.302.CLAD35` plugs and `EGG.00.302.CLL` sockets. Copper tubes with 2 mm outer diameter and 0.45 mm wall thickness are used as bend protection.

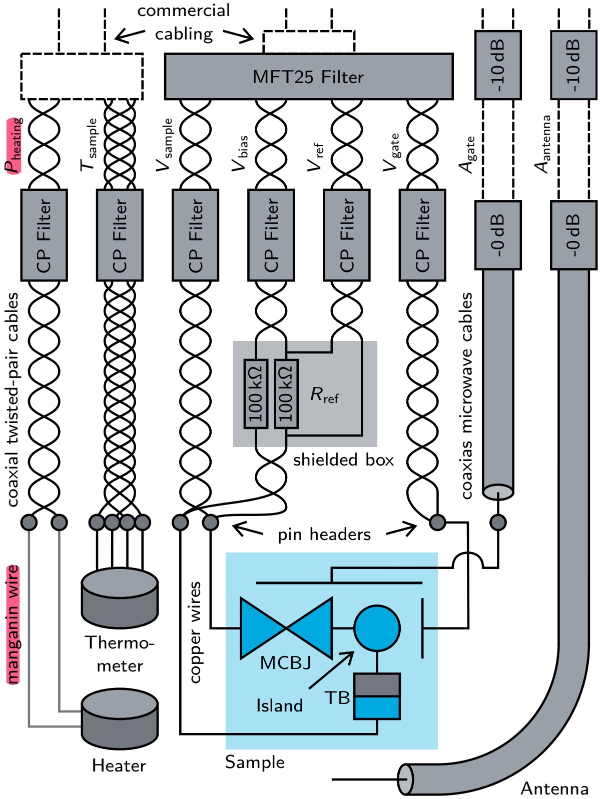


Figure 12 Custom cryostat cabling at the sample stage. All components shown are mounted at the base-temperature stage, except for the microwave attenuators at the 4 K stage, indicated by "-10 dB". The island is coupled to the MCBJ (left), the tunnel barrier (bottom), the DC gate (right), and the AC gate (top). Thin copper wires connect the sample pads to pin headers, which are connected either to low-frequency coaxial twisted-pair cables or to high-frequency coaxial cables. The cold reference resistors R_{ref} are mounted in an additional shielded box. Copper-powder (CP) filters and the commercial MFT25 filter suppress high-frequency leakage and thermalize the lines.

and outer conductors.

The base-temperature cables are soldered to pin headers. From there, 90 μm insulated copper wires connect to the sample pads through conductive silver paste. This short copper section gives a low contact resistance and provides a local thermal anchor close to the device. The voltage and current leads define a four-terminal measurement with respect to the cryostat wiring. They do not form an ideal Kelvin measurement of the atomic contact alone, because the local copper wires, silver-paste contacts, and sample pads remain inside the measured sample region. For the high-impedance devices considered here, this residual series resistance is negligible compared with the device resistance.

DC Filtering

The DC wiring carries the slowly varying bias and gate voltages as well as the voltage signals used to determine sample current and voltage. Its task is not only to pass low-frequency signals. It must also suppress high-frequency radiation that would heat the junction or create photon-assisted features unrelated to the intended microwave drive. The filter stack was therefore designed as both an electrical low-noise environment and a thermalization stage.

The low-frequency lines are filtered and thermalized in several stages. Four group-built copper-powder filters are mounted at the base-temperature stage, one each for the voltage, bias, current, and DC-gate lines. They follow the filter concept described in Ref. [26]. In this filter type, a copper signal wire is embedded in compacted copper powder inside a gold-plated copper housing. The filter has low resistance at DC and low frequency, but becomes lossy at high frequency. The relevant mechanism is the skin effect. Microwave currents are confined closer to the surface of the signal wire, while the surrounding copper powder provides a dissipative electromagnetic environment. The resulting attenuation is broadband and is not described by a single lumped RC cutoff.

The copper-powder filters act as a broadband protection stage against microwave leakage. At the same time, the large metal–metal contact area between wire, powder, and housing provides an efficient thermal anchor for the electronic degrees of freedom in the line. Their role in the present stack is to add high-frequency loss without adding a large DC resistance to the measurement circuit.

A commercial MFT25 microwave filter–thermalizer[9] is used in series with the copper-powder filters. According to the manufacturer, the MFT25 embeds the signal conductors in a highly conductive silver matrix that acts as a heat exchanger, and the compact package contains 25 filter lines in a gold-plated non-magnetic copper box. The manufacturer specifies strong microwave attenuation above 130 MHz together with low capacitance and good thermal contact to the millikelvin stage [27]. In the present setup, the MFT25 provides the compact

[9] Basel Precision Instruments, `MFT25-100Ohm SN0219`. The unit is connected through a Glenair, `25-way MWDM Micro-D` panel-mount receptacle.

commercial filter–thermalizer stage, while the copper-powder filters add additional broadband high-frequency loss.

The MFT25 also improved the practical reliability of the DC wiring. Earlier versions of the setup used fragile steel-capillary wiring, similar to the configuration described in Ref. [23]. These lines could break during assembly or thermal cycling, and inherited copper-powder filters failed intermittently. The MFT25 replaced most of this capillary-filter chain on the transport and DC-gate lines while adding a well-thermalized commercial filtering stage. The result was a mechanically more robust wiring system that was better suited to maintaining a low electronic temperature at the sample.

The reference resistors are mounted in the shielded base-temperature region close to the sample. In an earlier configuration they were located at the still stage. Moving them to the base-temperature stage reduces their Johnson–Nyquist voltage noise, whose voltage-noise spectral density scales as $\sqrt{T}$, and keeps the current readout in the same cold filtered environment as the sample [24, 25]. The design was also made mechanically more robust. An earlier through-hole metal-oxide resistor developed intermittent contacts, while the revised design uses an SMD resistor[10] on a small chip. This provides better thermal contact and a less fragile electrical connection. The reference-resistor region is additionally shielded, as indicated in Fig. 12, to reduce capacitive pickup from neighboring lines. Before cooldown, the wiring was checked by measuring the resistance matrix between the relevant ports and by verifying the ground connections of the lines.

The filtering stack was therefore optimized for low electronic temperature and robust wiring. In practice, the MFT25 and the revised reference-resistor design were as important for usable spectra as the nominal attenuation of the filters.

AC Cabling

In addition to the DC bias and gate lines, the setup contains two high-frequency paths for microwave excitation. I refer to them as the AC-gate line and the antenna line. The AC-gate line terminates on the on-chip AC gate, which is capacitively coupled to the sample. The antenna line is a stripped coaxial cable placed close to the sample. These names describe the hardware geometry, not a calibrated lumped circuit. At GHz frequencies, both paths can launch local electromagnetic fields that couple to the device region.

Both microwave paths are routed as coaxial lines. The commercial AC cabling inside the cryostat uses superconducting cables to keep the transmission high. At each temperature stage, the microwave line passes through a connector that also provides a thermalization point. The cryostat was initially delivered with attenuators installed at these positions. I replaced them by pass-through

[10] Scientific workshop, 012345678. Vishay/Draloric, `TNPU1206100KBZEN00`.

connectors[11] except at the 4 K flange. The remaining 10 dB attenuators at 4 K provide a defined thermalization point while placing the dissipated power at a stage with comparatively large cooling power. For future measurements, moving attenuation closer to the mixing-chamber stage would likely be preferable, because the electronic temperature at the device is more important than maximizing microwave transmission.

The base-temperature part of the microwave cabling[12] was installed together with Patrick Raif [7]. Frequencies up to about 20 GHz were transmitted reliably in the installed configuration, as inferred from microwave-induced changes in the spectra. The maximum applied source power was 10 dBm. In a matched 50 Ω system this would correspond to a peak voltage amplitude of about 1 V, but this number does not describe the local voltage at the junction [28]. The two paths also have different relative transmission as a function of frequency, which provides access to different effective power windows for different microwave tones.

Together with Patrick Raif, we showed that both paths act in practice like antennas for the device [7]. Part of the microwave power is launched already near the pin headers. From the point of view of the sample, the two paths therefore behave more like two different antennas than like two sharply separated drive mechanisms. The coupling to the sample is not impedance matched. A calibrated matched microwave environment would require a substantially different break-junction architecture, closer to the dedicated microwave design used by Bretheau [29]. I therefore did not attempt an absolute calibration of the local microwave voltage. Instead, I treat the microwave amplitude as an effective parameter and calibrate it only where the data provide an internal reference, for example through SIS photon-assisted tunneling as described in Sec. **??**. The reliable statements are the applied source settings, the relative response of the two paths, and the presence or absence of microwave-induced spectral features. The absolute local microwave amplitude at the junction remains unknown unless it is extracted from the data.

The microwave wiring therefore provides reproducible irradiation, but not a calibrated local voltage source. This distinction is central for the interpretation of microwave-driven maps. Frequency-dependent features can be analyzed directly, while source power is treated as an effective drive strength.

Thermometry and Heating

The sample space is equipped with dedicated thermometer and heater wiring to monitor and control the local sample-holder temperature. These lines are filtered by copper-powder filters, but they are routed through a separate cable tree and a separate break-out box. This keeps heater and thermometer signals away from the sensitive transport circuit. The separation is important because

[11] Amphenol SV Microwave, SF1597-6003.
[12] Elspec, DA50047-821, 50 Ω, CuAg/PTFE/Cu, 1.19 mm. Rosenberger, 02S141-270L5.

thermometer readout and heater operation can otherwise interfere with low-current measurements. The thermometer arrangement follows the earlier MCBJ setup used by Martin Prestel [23]. In the present setup, a ruthenium-oxide resistor thermometer[13] was mounted on the sample holder next to the sample.

The heater is a local resistive heater made from thin insulated manganin wire with a resistance of about 100 Ω. It is wrapped around one of the posts that connects the sample holder to the mechanical break-junction assembly. This geometry heats the sample holder locally while remaining mechanically simple and robust. I used the heater for controlled temperature-dependent measurements.

The thermometer reading should be interpreted as the local sample-holder temperature, not automatically as the electronic temperature of the junction. The electronic temperature can be higher because of imperfect line thermalization, residual high-frequency radiation, dissipated bias power, or microwave-induced heating. This distinction is particularly important under AC irradiation. A change in the measured temperature can indicate real heating, but it can also include direct microwave-induced readout artifacts.

The thermometer and heater lines are filtered and thermalized for the same reason as the transport lines. They connect room-temperature electronics to the millikelvin stage and can therefore act as thermal and electrical noise channels. Their voltage-noise requirements are less restrictive than those of the transport lines, but they still have to be well anchored to avoid parasitic heating. The heater cable[14] connects to the Bluefors cabling through a dedicated plug. I therefore treat the thermometer and heater wiring as part of the cryogenic wiring system, not as auxiliary cabling outside the measurement environment.

The temperature readout therefore provides a local sample-holder thermometer. Controlled heating and temperature-dependent spectra provide consistency checks, while the electronic temperature has to be inferred from transport data whenever it matters quantitatively.

Taken together, the cold electrical environment defines the sample-side measurement conditions. The cold reference resistors determine the current readout, the filtered and thermalized wiring limits electrical noise and electronic heating, and the microwave lines provide reproducible but not absolutely calibrated irradiation.

[13] Lake Shore Cryotronics Inc., `RX102B-CB`, serial number `U06030`, curve name `MCBJ Sample ORI`.
[14] LEMO, `FGG.0S.302.ZLAT`.

1.2.4 Instrumentation and Grounding

Figure 13 summarizes the room-temperature instruments and their grounding connections. This part of the setup had two tasks. It had to provide synchronized voltage readout and bias control, and it had to avoid additional ground paths that bypassed the filtered cryostat wiring. The cryostat was treated as the local ground reference of the experiment. This was practical because the cryostat is a large metallic body with a stable electrical potential, while the setup did not provide a separate measurement ground comparable to the P5 setup [4]. The working rule was to connect each instrument to the cryostat reference only through its intended signal path or through a deliberately isolated control link. Uncontrolled parallel connections between building ground, computer interfaces, instrument shields, and the cryostat body were avoided.

The control interfaces were therefore part of the grounding strategy, not merely a convenience layer. The control computer was connected to the laboratory network, while GPIB and RS-232 instruments were interfaced through USB adapters[15]. USB connections can introduce additional shield and ground paths, so they were isolated either by USB-over-LAN links[16] or by galvanic USB isolators[17]. The cryostat control unit[18] was included in this isolated control chain. LAN-connected instruments were used where available, because they allowed remote control without direct USB connections to the measurement computer. The motor controller and LAN switch[19] were floating in the final configuration because their plugs did not provide a protective-earth connection, as indicated in Fig. 13.

The voltage signals were first amplified by voltage preamplifiers[20]. These amplifiers provide selectable gains and a low-pass bandwidth of about 100 kHz. Their ground-referenced output voltages are digitized by the analog inputs of the measurement card[21]. Although the ADwin can sample at rates up to 100 kHz, the quasi-dc sweeps used here did not require this bandwidth. The data stream was therefore reduced on the ADwin to 137 Hz.

The acquisition strategy was to record synchronized voltage traces as directly as practical and to postpone non-essential filtering, smoothing, and gridding to the evaluation stage. This keeps irreversible processing out of the measurement chain. It is especially important when new measurement protocols are commissioned, because a processing mistake should remain correctable from the stored raw data rather than being built into the only copy of the spectrum. Once the voltage signals were digitized on the ADwin, they were therefore kept digital

[15] National Instruments, `GPIB-USB-HS`, used for GPIB access.

[16] Icron, `USB 2.0 Ranger 2304`.

[17] CESYS, `C028149`.

[18] Bluefors, `LD400`, serial number `BF2015-05`, connected by USB.

[19] D-Link, `DGS-108`.

[20] Femto, `DLPVA`, preamplifier. Scientific workshop, `WW-98000627`, power supply. Femto, `Luci10`, USB control interface.

[21] Jäger Computergesteuerte Messtechnik GmbH, `ADwin Gold II`, connected by LAN.

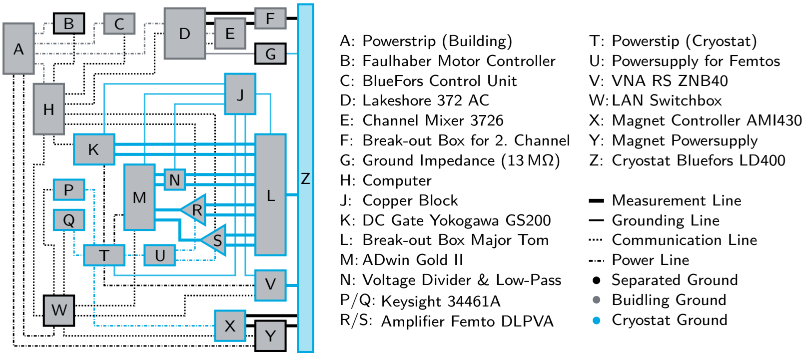


Figure 13 Grounding and instrument schematic of the room-temperature setup. Building ground and cryostat ground are kept separate except for a 13 MΩ connection (G) through the temperature controller. The motor controller (B) and LAN switch (W) are floating because their plugs do not provide a protective-earth connection. The magnet power supply (Y) was also operated floating, which minimized the observed noise in the spectra.

through storage and evaluation. Additional analog–digital, digital filtering, and digital–analog conversion stages inside intermediate instruments were avoided because they would obscure which operation changed the data and would make the final trace harder to audit.

The bias voltage was generated by an analog output of the same measurement card. Before entering the cryostat wiring, it passed through a warm symmetric voltage divider and a low-pass filter[22]. The divider reduced the full output range of the card to the voltage window relevant for the sample and improved the effective bias resolution. Its resistances were chosen low enough that their thermal noise and their contribution to the total source impedance remained limited.

The divider was kept at room temperature, in contrast to the cold shunt used in Ref. [23]. This preserved access to filter, wiring, and grounding diagnostics during cooldown and low-temperature operation, following the practical motivation of Ref. [4]. A shunt mounted inside the cryostat would have fixed part of the bias environment locally, but it would also have made several grounding and continuity checks inaccessible during a cooldown.

Several combinations of available voltage sources, amplifiers, and multimeters were tested before settling on the configuration shown in Fig. 13. The practical optimization criterion was the low-frequency noise in the sample- and reference-voltage channels under the same filtering and grounding conditions used for transport measurements. The relevant observable was not only electrical conti-

[22] Thorlabs Inc., EF120.

nuity, but the stability and cleanliness of the measured spectra.

Using one measurement card for both voltage channels ensured that the sample-voltage and reference-voltage readings were acquired as synchronized pairs. During a typical measurement, the bias was applied symmetrically around ground so that the common-mode voltage of the sample remained close to zero. The voltage was ramped slowly down and up, with full periods of 5 min or longer.

The remaining instruments provided gate control, microwave excitation, temperature control, mechanical actuation, magnetic field control, and diagnostics. The DC gate voltage was applied with respect to ground by a high-precision voltage source[23]. All DC lines were routed through a break-out box[24] into the commercial cryostat cabling. The microwave excitation paths were driven by the two ports of a vector network analyzer[25], which were connected to the AC-gate and antenna lines.

Temperature monitoring and local heating were handled by an AC resistance bridge[26]. It read out the sample heater, the sample thermometer, and auxiliary thermometers through the separate thermometry cable tree. I replaced the previously used AC resistance bridge because the newer unit can continuously read the sample thermometer while the second channel, together with the channel scanner, reads the auxiliary thermometers. This required an additional break-out box[27] to separate the dedicated sample-thermometer line from the auxiliary thermometry lines. The continuous sample-temperature readout allowed PID control of the local sample temperature to a chosen setpoint.

Mechanical actuation of the break junction was provided by an encoded motor and driver[28]. A one-axis 7 T magnet supply[29] was used when superconductivity had to be suppressed. Additional diagnostic access was provided by a precision source and several digital multimeters[30]. These diagnostic instruments were connected only when their information was needed, because each additional instrument could introduce another pickup or ground path.

For the lowest-noise spectra, devices that were not required during the actual sweep were disconnected from the setup. In particular, the magnet controller and magnet power supply were usually disconnected when not in use. This mattered because the magnet supply is a large high-current instrument and can couple low-frequency pickup or ground offsets into the cryostat environment even when the magnetic field is constant. The communication cables of the preamplifiers were removed whenever remote gain control was not needed. This reduced the number of possible ground-loop and pickup paths connected to the sensitive

[23] Yokogawa, `GS200`, connected by GPIB.
[24] Scientific workshop, `WW-2280090`, "Major Tom".
[25] Rohde & Schwarz, `ZNB40`, connected by LAN.
[26] Lake Shore Cryotronics Inc., `372AC` and `3726`, connected by USB.
[27] Scientific workshop, `WW-2480259`, thermometry break-out box.
[28] Faulhaber, Motor `2232R024S`, Encoder `R IE2-1024 4613`, Driver `MCDC3002S RS`, Gear Head `22/2S`, connected by RS-232.
[29] American Magnetics Inc., `AMI430`, connected by LAN.
[30] Keysight, `B2962A`, connected by LAN. Keysight, `34461A`, connected by LAN.

voltage-readout circuit.

The final grounding configuration was therefore selected empirically. It preserved a well-defined cryostat reference while minimizing pickup in the measured voltage channels. It should be understood as the working grounding solution for this specific cryostat, instrument rack, and wiring configuration. It is not a universal grounding model, but it documents which choices produced stable spectra in this experiment.

Taken together, the instruments and grounding define the room-temperature side of the measurement environment. The acquisition card pairs the voltage traces, the bias chain sets the control signal, and the grounding scheme suppresses low-frequency pickup. These constraints are part of the interpretation of the transport curves discussed in the following chapters.

1.3 Digital Infrastructure

The transport measurements in this thesis depend on more than the electrical circuit described above. Each spectrum also has to remain connected to the state of the surrounding experiment: amplifier gains, reference resistance, bias waveform, temperature, microwave settings, magnetic field, gate voltage, and the mechanical position of the break junction. Without this context, a recorded voltage trace would not be a reproducible physical measurement, but only a time series from the acquisition card.

The digital infrastructure was therefore treated as part of the experimental apparatus. Its role was to connect instrument control, synchronized acquisition, metadata logging, and data evaluation into one traceable path from raw voltages to calibrated *I*–*V* curves and fitted model parameters. This section first describes the measurement architecture used to acquire and store the data. It then describes the evaluation layer that converts raw HDF5 records into the common data products used for tunnel-regime BCS analysis and finite-transmission MAR fitting.

No Settings manually. just digitally and reproducable

1.3.1 Digital Measurement Architecture

The measurements in this work were controlled by a layered Python infrastructure. The first layer, `p5control`, provided the general abstraction for instrument access, server-based control, and HDF5 data storage. The second layer, `p5control-bluefors`, adapted this framework to the Bluefors break-junction setup by defining the relevant drivers, calculated diagnostic channels, and spectroscopy protocols. Together, these layers connected the physical instruments described in Sec. 1.2.4 to the raw voltage traces and metadata used for the later data evaluation.

`p5control`: Instrument Abstraction and Data Acquisition

The experiment required a control layer that could handle many slow instruments during quasi-dc spectroscopy without turning each measurement script into a collection of device-specific commands. The task of this layer was to keep instrument control, data acquisition, and metadata logging in one reproducible structure.

The general framework for this task was `p5control`, which was written by Florian Kayatz during his student-assistant work under my supervision [30]. The central idea of the framework is to separate device-specific communication from measurement scripts and graphical user interfaces. Physical instruments are represented by drivers. These drivers expose the relevant control parameters and readout values in a common form, while the details of LAN, GPIB, RS-

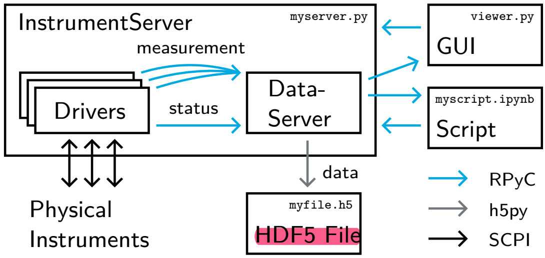


Figure 14 Architecture of the `p5control` measurement framework. Device-specific drivers connect physical instruments to an `InstrumentServer`; a separate `DataServer` writes status and measurement data to an HDF5 file. Scripts and graphical user interfaces communicate with both servers through RPyC.

232, USB, or device-specific command syntax remain hidden inside the driver implementation.

Figure 14 shows the resulting architecture. The `InstrumentServer` and `DataServer` separate instrument control from file writing, while measurement scripts and graphical user interfaces communicate with both servers through RPyC [31]. This made the same instrument layer available for interactive control widgets and scripted measurement protocols, and it kept the file-writing logic independent of individual measurement scripts.

During operation, `p5control` distinguishes between status data and measurement data. Status data are recorded continuously. They document the slowly changing state of the setup, for example temperatures, motor position, magnet field, microwave settings, amplifier gains, or output states. Measurement data are written when a measurement is started. They contain the time traces acquired for a specific sweep or protocol. This distinction is important because the physical context of a spectrum can change on a slower time scale than the spectrum itself.

The data are stored in an HDF5 file using `h5py` and the HDF5 data model, as sketched in Fig. 15 [32, 33]. The file contains both the raw traces used for later evaluation and the surrounding instrument state needed to interpret them. It therefore acts as a measurement record and as a digital laboratory notebook.

The framework was designed for reproducible spectroscopy rather than hard real-time feedback. Timing-critical acquisition remained on the ADwin, which recorded the synchronized voltage channels during a sweep. `p5control` coordinated the higher-level experiment: it configured instruments, started measurements, collected data streams, and wrote the resulting traces and metadata to disk. This division matched the requirements of the break-junction experiment, where individual spectra were slow, but the relation between raw voltage traces and setup state had to remain reproducible. It made each spectrum traceable to synchronized voltage traces, instrument settings, and the logged state of the

```
measurement/mpath0/dev0
                   dev1
            mpath1/dev0
                   dev1
     status/dev0
            dev1
```

(a) myfile.h5

t	x1	x2
0.00	0.12	3.45
0.01	1.23	4.56
0.02	2.34	5.67
0.03	3.45	6.78
...	...	...

(b) .../mpath0/dev0

t	a	b
0.0	0.5	0.1
1.0	1.0	0.1
2.0	1.0	0.2
3.0	0.0	0.2
...	...	...

(c) status/dev0

Figure 15 HDF5 data structure used by `p5control`. The `measurement` branch stores sweep-specific datasets, while the `status` branch stores continuously logged setup quantities. Both are written as time-indexed tables.

setup.

`p5control-bluefors`: Setup-Specific Control Layer

The generic `p5control` layer provided the server structure, data storage, and driver abstraction, but it did not define the measurement logic of the Bluefors break-junction setup. This setup required a spectroscopy-oriented control layer that knew which instruments belonged to one transport measurement and which quantities had to be stored with each spectrum. I implemented this setup-specific layer in `p5control-bluefors`, including device-specific drivers, graphical control widgets, calculated transport channels, and the measurement protocols used for the data in this thesis [34].

The layer treats each controlled object through the same practical model, summarized in Fig. 16: parameters, values, and state. Parameters are quantities set by the user or by a measurement script, for example bias amplitude, amplifier gain, microwave source settings, temperature setpoint, magnetic-field target, or motor displacement. Values are quantities read back from instruments or calculated from measured channels. State records whether a device is actively sweeping, irradiating, heating, ramping, moving, outputting, or reporting a diagnostic condition such as amplifier overload. This model kept slow control variables and safety-relevant state changes in the same status log as the measured spectra.

The primary measurement record consists of the synchronized voltage traces recorded by the ADwin and the instrument state stored around the sweep. Together with the selected Femto gains and the reference resistance, these raw voltage traces define the later reconstruction of V_{sample} and I_{sample}. In `p5control-bluefors`, the same conversion was also implemented in the `Calc` driver as a diagnostic layer for live monitoring. This layer was useful during setup work and interactive checks, but it was usually inactive during actual measurements. The spectra used for analysis were derived separately from the stored raw voltage traces, as described in the following subsection. The remaining drivers

driver	parameters	values	state
ADwin	V_0, T_0, f_0	V_1, V_2, i_0	Output, Sweeping
Femtos	A_1, A_2		Overload (2)
VNA	A_{out}, ν	S_{11}, S_{21}	Irradiating
BlueFors	P_{heater}	T_{sample}, $(T, p)_{\text{aux}}$	Heating
Yoko	V_{gate}	I_{leak}	Output
Magnet	$\mu_0 H$, $\frac{\partial}{\partial t}\mu_0 H$	$\mu_0 H$	Ramping
Motor	Δx, $\frac{\partial}{\partial t}\Delta x$	Δx	Moving
Calc	R_{ref}	$(I, V, G, R)_{\text{sample}}$	Calculating
		$(V_1, V_2)_{\text{offset}}$	

Figure 16 Setup-specific driver model used by `p5control-bluefors`. Each driver exposes set parameters, read or calculated values, and state variables. Parameters and states are logged as status data, while measurement values are stored with the corresponding spectrum. Gray entries mark diagnostic quantities.

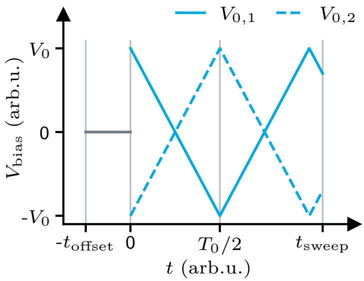
provide the physical context of the spectrum: microwave amplitude and frequency, sample temperature, displacement of the break junction, magnetic field, and gate voltage. In this sense, `p5control-bluefors` connects the electrical reconstruction described in Sec. 1.2.3 to the structured data record described above.

This connection is enforced by the measurement protocol shown in Fig. 17. A single *I*–*V* spectrum is not only a voltage ramp. The relevant gains, reference value, bias waveform, timing parameters, and instrument states have to be fixed before the sweep, and the synchronized voltage traces have to be stored together with these settings. The script `single_iv.py` therefore defines one complete spectrum: it sets the sweep parameters, waits through an offset interval, records the sweep, and writes the acquired data and metadata to the HDF5 file.

Parameter-dependent measurements are built by repeating this elementary spectrum. The script `iv_script.py` steps through an external parameter axis, for example microwave amplitude, frequency, sample temperature, break-junction displacement, magnetic field, or gate voltage. For each point it starts one `single_iv.py` measurement and stores the result in the corresponding measurement path. This structure makes maps reproducible because each line or point in the map is linked to the raw voltage traces and to the status of the surrounding setup at the time of acquisition.

1.3.2 Data Evaluation with `superconductivity`

The raw HDF5 files produced by the measurement stack contain synchronized instrument outputs and setup metadata, but they are not yet the physical data products used in the analysis. This separation is intentional: the measurement stores the least processed useful record, while filtering, offset cor-



(a) Sequence of single *I*–*V* measurement

`single_iv.py`	
(V_0, T_0):	sweep (amplitude, period)
$(t_{\rm offset}, t_{\rm sweep})$:	time of (offset, sweep)
(A_1, A_2):	channel amplifications
f_0:	sampling rate
`iv_script.py`	
$(A_{\rm out}, \nu)$:	microwave irradiation
$T_{\rm sample}$:	sample temperature
Δx:	sample displacement
$\mu_0 H$:	magnetic field
$V_{\rm gate}$:	gate voltage

(b) Settings of measurement sequences

Figure 17 Protocol for quasi-dc *I*–*V* spectroscopy. (a) A single spectrum consists of an offset interval followed by a controlled bias sweep. (b) `single_iv.py` defines the sweep settings, gains, and sampling rate. `iv_script.py` repeats this spectrum over an external parameter axis.

rection, gridding, and smoothing are applied later with explicit parameters. I first implemented many of the basic evaluation ideas in the setup-specific repository `p5control-bluefors-evaluation` [35]. For this thesis I then developed the broader Python package `superconductivity` [36] as the evaluation layer between acquisition and interpretation. In addition to trace evaluation, `superconductivity` contains numerical models for BCS and MAR spectra, fitting routines, a graphical user interface, and a visualization pipeline. The graphical interface and visualization tools were useful for daily work but are not discussed in detail here. This subsection focuses on the reproducible data path from raw voltage traces to calibrated *I*–*V* spectra, uniformly sampled grids, and fitted model parameters.

Evaluation

The first evaluation step converts one recorded measurement path into one raw *I*–*V* trace. The ADwin stores the synchronized channels V_1, V_2, and i_0 on the same time base. The voltages V_1 and V_2 are the amplified room-temperature output signals of the two readout channels. They are therefore not yet the sample voltage and current. The additional channel i_0 records the internal trigger state of the ADwin program and marks which part of the measurement sequence a point belongs to.

I use i_0 to split the recorded trace into the offset interval and the individual sweep branches. This avoids inferring sweep boundaries only from the noisy voltage ramp. For each spectrum, the evaluation selects either the upward or downward branch and discards the other parts of the sequence. Unless stated otherwise, the spectra shown in this thesis use the downward branch. This gives one monotonic sweep segment with synchronized values of $V_1(t)$ and $V_2(t)$.

The selected branch is then converted from amplifier output voltages to the physical transport variables. With voltage gains A_1 and A_2, reference resistance R_{ref}, and channel offsets $V_{1,\mathrm{off}}$ and $V_{2,\mathrm{off}}$, the raw sample voltage and current are

$$V(t) = \frac{V_1(t) - V_{1,\mathrm{off}}}{A_1}\,, \tag{10}$$

$$I(t) = \frac{V_2(t) - V_{2,\mathrm{off}}}{A_2 R_{\mathrm{ref}}}\,. \tag{11}$$

This coarse offset subtraction is performed for every spectrum. It uses the offset interval recorded before the sweep and compensates slow thermal drifts of the Femto amplifier outputs. A second, smaller offset correction can be applied later as an optional fine tuning step when the symmetry-based offset analysis described below is used. At this point the data are still an ordered time trace, not a uniformly sampled function of voltage or current.

The raw branch is not yet suitable for numerical derivatives or direct comparison between spectra. Voltage and current are recorded as time traces, while the later analysis treats each spectrum as a function of sample voltage or current. I therefore convert each selected branch to a regular grid. Figure 18 illustrates the steps for one representative trace.

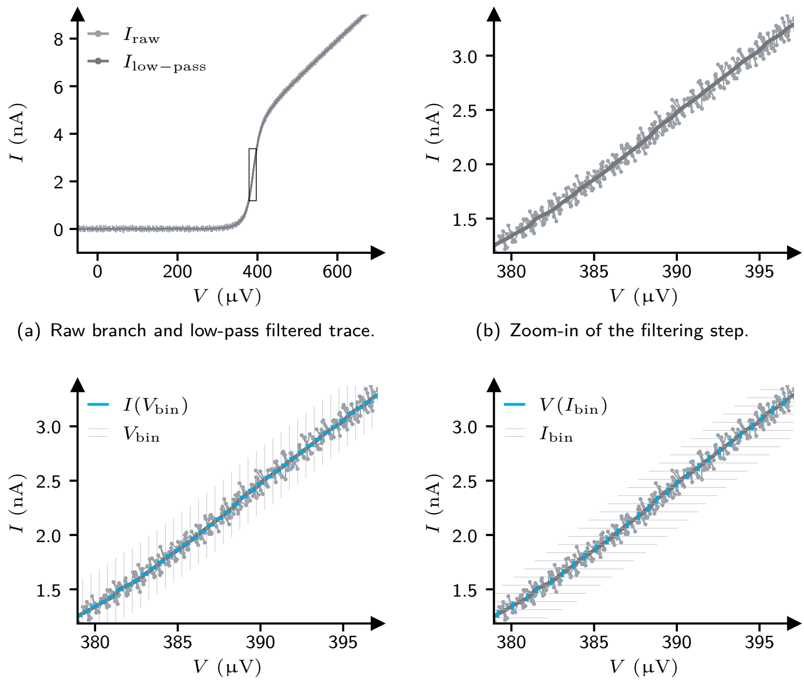
The branch entering the gridding pipeline has therefore already been corrected by the offsets introduced in Eq. (11). If an additional offset analysis is performed, the resulting fine correction is applied at this stage before filtering and binning. A low-pass filter can then suppress high-frequency pickup that is not resolved by the quasi-dc sweep. This filtering is applied to the selected branch, not to the full measurement sequence, so that the discontinuities between offset interval and sweep do not enter the filtered trace.

The filtering and gridding parameters were kept explicit in the evaluation record. The standard analysis preset used a fourth-order Butterworth low-pass filter with a $43.7\,\mathrm{Hz}$ cutoff and resampled the selected branch onto a uniform time axis at the same rate before binning. Typical voltage grids covered $-1.6\,\mathrm{mV}$ to $1.6\,\mathrm{mV}$ with 1601 points, corresponding to $2\,\mu\mathrm{V}$ spacing. Typical current grids covered $-30\,\mathrm{nA}$ to $30\,\mathrm{nA}$ with 2001 points, corresponding to $30\,\mathrm{pA}$ spacing. After binning, the default smoothing consisted of a three-bin median filter followed by Gaussian smoothing with $\sigma = 2$ bins.

This resampling step does not add physical information. Its purpose is to avoid empty or sparsely populated bins when the sweep is later projected onto a voltage or current axis. The data are then assigned to equally spaced bins. For a voltage-based representation, the current values in the bin around V_k are averaged,

$$I(V_k) = \langle I(t_j) \rangle_{V(t_j)\in[V_k-\Delta V/2,\,V_k+\Delta V/2]}\,. \tag{12}$$

The same operation can be performed with the axes exchanged to obtain $V(I_k)$. This second representation is useful for branches with steep current changes,



(a) Raw branch and low-pass filtered trace.

(b) Zoom-in of the filtering step.

(c) Voltage-grid representation $I(V)$

(d) Current-grid representation $V(I)$

Figure 18 Evaluation pipeline for one selected sweep branch. The trace is low-pass filtered, resampled to increase the point density, and binned onto a regular voltage or current grid.

where a voltage grid would concentrate many points in a small voltage interval. Optional smoothing is applied only after binning, so that derivatives are taken from a regular data product rather than from the irregular raw trace.

This pipeline yields the evaluated *I–V* curve used for plotting, numerical differentiation, and model comparison. Its output remains tied to one selected sweep branch and to the acquisition metadata described above.

The evaluated spectra form the common interface between acquisition and the model-dependent parts of the thesis. In the tunnel regime, the same calibrated *I–V* or *dI–dV* data can be compared to BCS-Dynes quasiparticle spectra and photon-assisted tunneling models. In atomic contacts, the finite-transmission spectra are compared to MAR calculations. This separation keeps the trace treatment independent of the physical model, while the later fits operate on a reproducible data product with fixed units, grids, offsets, and metadata.

Offset Analysis

The coarse offset subtraction described above removes the dominant amplifier drift. For quantitative spectra, I could additionally apply a smaller symmetry-based correction to align the residual voltage origin. This step was optional and was only used when the measured curve contained enough symmetric structure to determine the remaining offset reliably. The standard voltage-offset scan covered $-45\,\mu\mathrm{V}$ to $45\,\mu\mathrm{V}$ in $0.2\,\mu\mathrm{V}$ steps. The analogous current-offset scan covered $-0.35\,\mathrm{nA}$ to $0.35\,\mathrm{nA}$ in $1\,\mathrm{pA}$ steps.

The voltage-offset search is based on a correlation-like alignment of the differential conductance with its mirrored counterpart. The method assumes that the intrinsic quasi-dc *I*–*V* curve is point symmetric. For such a curve, $\mathrm{d}I/\mathrm{d}V$ is even in voltage. This is useful because a constant current offset drops out under differentiation. The remaining problem is therefore the horizontal alignment of the conductance trace. If the correct voltage origin is chosen, the derivative overlaps with its reflection. In the implementation, this idea is written as a mirror-symmetry error rather than as a product-type correlation function.

In practice, the analysis scans a list of candidate voltage offsets. For a candidate δV, the trace is shifted by subtracting δV, then passed through the same filtering, resampling, binning, and smoothing pipeline as the final data. The shifted trace defines a sampled curve $I_{\delta V}(V_k)$ on a symmetric voltage grid. The score is the mean absolute mirror mismatch of its differential conductance,

$$\epsilon_V(\delta V) = \left\langle \left| \frac{\mathrm{d}I_{\delta V}}{\mathrm{d}V}(V_k) - \frac{\mathrm{d}I_{\delta V}}{\mathrm{d}V}(-V_k) \right| \right\rangle_k . \tag{13}$$

The voltage offset assigned to the trace is the candidate with the smallest finite value of ϵ_V.

The procedure is a diagnostic correction, not a physical model fit. It is meaningful only when the expected *I*–*V* curve is point symmetric over the analyzed bias window. Real asymmetries, rectification, strong hysteresis, or too narrow a bias range would bias the minimum of Eq. (13). The same principle can be applied to determine a residual current offset by exchanging the axes and comparing $\mathrm{d}V/\mathrm{d}I$ with its mirrored copy.

BCS Simulation and Fitting

The BCS simulation implements the forward model used for tunnel-regime quasiparticle spectra. The microscopic tunnel expressions were introduced in Sec. **??**. Here I describe how they are evaluated numerically and used in fits.

The core model evaluates the SIS tunnel current from Eq. (**??**). The two electrodes are represented by Dynes-broadened BCS densities of states and finite-temperature Fermi functions. The free or fixed model parameters are the normal conductance G_N, electronic temperature T, zero-temperature superconducting gap Δ_0, and Dynes broadening γ.

For repeated evaluations, the production model uses the convolution form of the tunnel current, Eq. (**??**). Instead of recomputing the full energy integral for every bias point, the occupied and empty spectral functions are correlated on an energy grid and interpolated back to the requested voltage axis. This makes parameter sweeps and fits substantially faster while keeping the same physical model.

Photon-assisted tunneling is included by applying the Tien–Gordon sideband sum of Eq. (**??**) to the BCS current. This extension adds the irradiation amplitude A and frequency ν as model parameters. The model can also include an optional Gaussian voltage-noise broadening, which averages the current over a distribution of bias offsets with width σ_V.

The fitting interface was designed so that each model parameter can either be optimized or kept fixed. This is important for map data, where global parameters such as Δ, γ, G_{N}, ν, or σ_V may be determined independently or from a reference trace and then held constant while only T or A is fitted for each spectrum. Fixing well-constrained parameters reduces correlations between fit parameters and makes the extracted trends easier to interpret.

The JAX backend is used for production fits because the model is evaluated many times for each spectrum and for complete maps. JAX provides composable transformations for just-in-time compilation and vectorized array evaluation, which makes it well suited for repeated numerical model evaluation [37]. Computational cost was treated as part of the evaluation design: JIT compilation, vectorized array operations, and the convolution kernel keep typical fitting tasks on the order of minutes on a standard laptop. The NumPy implementation serves as a transparent reference backend.

The resulting fits should be interpreted as tunnel-regime quasiparticle fits. They extract effective parameters within the BCS-Dynes description, but they do not describe finite-transmission MAR, coherent supercurrent dynamics, Coulomb blockade, or environmental phase fluctuations.

MAR Simulation

The MAR simulation implements the finite-transmission forward model used for atomic-contact spectra. The physical model is described in Sec. **??**; here I only describe the numerical layer used for evaluation. The core calculations were based on three compiled Fortran backends from Carlos Cuevas: a symmetric Hamiltonian approach solver, an asymmetric Hamiltonian approach solver, and a full-counting-statistics solver. The two Hamiltonian approach solvers evaluate the dc MAR current described in Ref. [38], while the FCS backend gives the charge-resolved decomposition of Ref. [39]. I did not modify these numerical kernels.

My contribution was the shared Python layer around all three backends. This layer handled units, model dispatch, temperature-dependent gaps, deterministic

HDF5 caching, odd-symmetry reconstruction, segmented backend calls, and parallel evaluation. The user-facing parameter was the zero-temperature gap, while the backend was called with the temperature-dependent gap. This was important when temperature was scanned or fitted, because a change in T affects both the Fermi occupations and the MAR voltage scale.

The reason for this additional layer was computational cost. In my evaluation environment, an uncached single-channel calculation on a sufficiently resolved voltage grid typically took a few tens of seconds with the symmetric Hamiltonian approach backend, about one to two minutes with the asymmetric backend, and about ten minutes with the FCS backend. Dense parameter scans would therefore be impractical if every voltage grid and parameter point were evaluated from scratch. The wrapper reduced this cost without changing the physical model: symmetric electrodes were sent to the faster symmetric backend, asymmetric electrodes to the asymmetric backend, and charge-resolved calculations to the FCS backend only when needed. Cached curves were stored under deterministic keys built from the rounded model parameters and a quantized voltage axis. Since the dc *I*–*V* curve is point symmetric, only positive voltages were evaluated and cached, while the negative branch was reconstructed.

MAR Fitting

To determine the pincode of an atomic contact, the measured finite-transmission spectrum has to be compared to MAR theory. The fit parameters for a multi-channel contact were therefore the channel transmissions τ_i and, depending on the data set, the effective temperature T, the zero-temperature gap Δ_0, and the broadening γ.

For fitting, the experimental trace was first mapped onto the non-uniform voltage grid of the theory landscape. This grid used many points in the sub-gap region, where the channel transmissions determine the MAR structure, and fewer points far outside the gap, where the spectrum mainly constrains the total transmission. Only voltage points for which both experimental and theoretical values existed were kept. I compared the traces using the normalized finite-bias conductance. This avoids numerical differentiation of noisy MAR traces while still comparing the nonlinear subgap structure. An optional weight vector could mask voltage regions that should not constrain the MAR fit, most importantly a central switching or supercurrent feature that is not described by the dc quasi-particle MAR model.

The actual fit was a grid-based landscape search. Candidate pincodes were generated from the discrete transmission grid and restricted to a plausible interval of total transmission, $\tau_{\Sigma,\min} \leq \sum_i \tau_i \leq \tau_{\Sigma,\max}$. For each candidate and for each point in the (T, Δ, γ) grid, the residual score was

$$\chi^2(\boldsymbol{\tau}, T, \Delta, \gamma) = \sum_{k \in \mathcal{M}} w_k \left[G_{\mathrm{exp}}(V_k) - \sum_i G_{\mathrm{HA}}(V_k; \tau_i, T, \Delta, \gamma) \right]^2 , \tag{14}$$

where $\mathcal{M}$ denotes the masked common voltage grid. The expensive part of this comparison was vectorized and JIT compiled with JAX [37]. In practice, preparing the cached theory landscape was the time-consuming step and took on the order of a few hours on a standard laptop for the grids used here. Once this cache existed, fitting an individual spectrum only required the landscape comparison and typically ran on the order of a few tens of seconds. The resulting landscape was converted into weights proportional to $\exp\bigl(-\chi^2/2\bigr)$, from which the best global parameter set and the channel-resolved transmission weights were extracted. The final channel transmissions were obtained from the maxima of these weights, with Gaussian peak fits used as a smooth interpolation on the discrete transmission grid.

This fitting procedure should be interpreted as a controlled numerical approximation to a continuous MAR fit. It assumes that the residual landscape is sufficiently smooth on the chosen parameter grid and that the physical minimum lies inside the scanned parameter ranges. The extracted transmissions are therefore effective channel transmissions within the coherent short-contact MAR model. The procedure does not model the dc supercurrent branch itself, environmental phase fluctuations, Coulomb blockade, or nonequilibrium heating beyond the effective parameters included in the grid.

1.4 Practical Notes on Sample Fabrication

This section collects practical notes from the laboratory that complement the conceptual description in the main text. It is not intended as a fully standardized protocol. Instead, it documents the small but critical steps, common pitfalls, and empirically useful habits that make reliable sample preparation more likely.

Wafer Preparation

Start with a 300 µm thick bronze wafer, which is typically covered by a protective foil. To remove this foil without damaging the wafer, cool the wafer in liquid nitrogen and then gently peel off the brittle fragments. This method has proven effective and largely non-destructive. The wafer is then polished. For this purpose, attach a polishing head made of sewn cotton cloth to a drill and use aluminum oxide from a white polishing compound block as the abrasive. The polishing head should be well moistened, preferably soaked, with a solvent such as ethanol or IPA. Fix the wafer to a smooth milled steel block using double-sided adhesive tape[31], taking care that no tape protrudes beyond the wafer edge. During polishing, move the polishing head from top to bottom so that debris generated in the process is less likely to settle back onto the wafer surface. Rotate the steel block periodically, since the lower part of the wafer is generally easier to polish uniformly. Polishing is complete once the entire surface shows a homogeneous mirror-like finish. To detach the wafer from the steel block, cool it again in liquid nitrogen. For cleaning, rinse the wafer first with acetone and then with IPA, ensuring that the acetone is removed completely before switching solvents. Finally, dry the wafer with pressurized nitrogen while the IPA is still wet in order to avoid residue formation. [40]

An alternative polishing method uses sandpaper with progressively finer grit on a glass slide, as described by Patrick Raif [41]. It is better suited to reducing long-range surface variations, but it is also considerably more time-consuming and was therefore not used routinely.

Next, prepare the spin coater, the polyamide (PA), and the vacuum oven. The PA is pre-portioned into small crimp-seal vials to minimize moisture exposure and to reduce the time required for warming. Remove one vial from the freezer and allow it to reach room temperature while preheating the convection oven to 135 °C. Line the inside of the spin coater with aluminum foil and verify that the vacuum oven is operational and open.

To remove adsorbed water, place the wafer on a hot plate at 100 °C for at least one minute. For reliable adhesion to the 1 in chuck, apply a layer of parafilm and bend it securely around the chuck edge. Then make five small holes in the parafilm using a syringe: one in the center and one near the edge in each cardinal

[31] e.g. Tesa Universal

direction. Allow the wafer to cool to room temperature before centering it on the chuck, and only then switch on the vacuum pump. Before dispensing the PA, it is advisable to run the spin-coating program once as a quick check that the wafer is held securely. This is particularly important for larger wafers, where insufficient adhesion may not be immediately obvious.

Blow the wafer dry with nitrogen to remove residual dust. Pour the PA onto the wafer so that roughly 90 % of the surface is covered, taking care to avoid trapped air bubbles. If bubbles remain, move or burst them carefully with a syringe without scratching the wafer. The spin-coating program consists of an initial spread step at 300 rpm for 30 s followed by a final spin step at 5000 rpm for 90 s. Immediately afterward, transfer the wafer to the convection oven for a 5 min soft bake at 135 °C. The hard bake is then performed in the vacuum oven using a programmed cycle that ramps to 400 °C, holds for 30 min, and cools back to room temperature over several hours. Detailed instructions and the corresponding temperature profile are given in Ref. [41]. In practice, starting the bake immediately after spin coating has been found to improve reproducibility.

The application of the EL11 and A4 resist layers follows the same general workflow, but the individual spin and baking parameters differ. In this case, aluminum foil inside the spin coater is unnecessary because these resists can be removed easily with acetone. The age of the resist should be checked before use, since prolonged storage can noticeably degrade performance. In particular, EL11 is first followed by its soft-bake step. After the A4 layer is applied, the subsequent A4 baking step also hard-bakes the underlying EL11 layer. The relevant process parameters in this subsection follow Table 2.

The wafer is then cut into smaller pieces of 3×8 mm using a custom shear-sheet cutter equipped with 3 mm and 18 mm spacers. Before cutting, clean the cutting edges thoroughly with IPA to reduce contamination and to maintain clean cuts. Samples taken from the outer wafer region are more likely to show thickness variations in the PI and resist layers. For reproducible fabrication, pieces from the central region of the wafer are therefore preferred. Edge-related thickness variations can affect the dose required during electron-beam lithography. After cutting, store the samples in a chip tray[32] for safe handling.

Electron beam lithography

The next step is electron-beam lithography (EBL) using the Crossbeam system. Begin by measuring the beam current with the jig containing the Faraday cup. A position list in the Zeiss software is helpful for locating the Faraday cup quickly. For reliable operation, the working distance should be kept as close as possible to 5.00 mm. Since the beam current may require some time to stabilize, it is often useful either to wait before measuring or to return later and repeat the check.

Accurate horizontal alignment of the sample is essential for exposure. The

[32] e.g. Entegris

alignment tools in the Focused Ion Beam (FIB) toolbox simplify this step. A practical approach is to zoom out to the maximum field of view and use the upper-right corner of the sample together with the farthest left sample edge for the initial alignment. This is usually sufficient to correct the dominant angular misalignment. Additional refinement within Elphy Plus is possible, but in practice it rarely improves the final angular alignment once the sample is mounted in the cryostat.

Next, focus the beam in the upper-right corner of the sample. Use the wobble function to center the beam in the aperture and minimize astigmatism with the x- and y-stigmators. Once the beam is well focused, check the current again at the Faraday cup. For the 30 μm aperture in high-current mode, the beam current should be about 0.5 nA. This setting is used for the small high-resolution structures. For the 120 μm aperture, which is used for the 1000 μm writing fields, the exact current is less critical because the larger leads tolerate moderate underexposure. In practice, a slight underexposure in these larger fields can even reduce writing time without compromising the result. The beam current for the 120 μm aperture in high-current mode is typically around 5 nA. Perform a final focusing step at the central position along the upper sample edge. A dirt particle on the resist is often useful as a focusing aid. At this stage, confirm once more that the working distance is 5.00(1) mm and that the beam shift is set to zero.

Before starting the exposure, note the current time. Since the design layout includes a timestamp, this provides a convenient way to match laboratory notes to the physical sample. If the measured beam current differs from earlier sessions, recalculate the settling time for the 100 μm field accordingly. The executed position list in Elphy Plus then locates the writing fields automatically, applies the corresponding settings, and writes the design. Stitching between adjacent writing fields is enforced by overlapping their nominal positions by 100 μm.

In practice, it is advantageous to keep the written design as simple as possible and to avoid an unnecessarily large number of polygons. This reduces writing overhead and makes the exposure procedure more robust. In addition, contamination dots were deliberately avoided as focusing markers. Instead, the combination of high-current mode and the 30 μm aperture proved sufficient for reliable focusing because it provides a comparatively large depth of focus. The drawback is that very low exposure doses are no longer readily accessible, but this was not a practical limitation for the structures used here.

Repeat the full alignment, focusing, and writing sequence for the remaining samples until a batch of roughly four to five samples has been completed.

If several weeks have passed since the previous EBL session, it is advisable to verify the alignment between the 100 μm and 1000 μm writing fields before exposing the first sample. A simple check is to expose one 100 μm field together with a neighboring 1000 μm field in the upper-right corner of a sample. The exposed resist appears visibly different from the unexposed surroundings and can therefore be used to judge the relative alignment. Prolonged observation should be avoided, however, since the resist can be exposed unintentionally and lose contrast. If misalignment is found, adjust the settings accordingly. For example, if the 100 μm field is shifted to the left, correct U by $-\Delta X$; if it is shifted upward, correct V by $-\Delta Y$. Repeat the test until the alignment is satisfactory.

After exposure, the samples are developed. Prepare a crimp-seal vial containing a mixture of one part MIBK and three parts IPA, and prepare one additional vial of pure IPA for each sample. Immerse the sample in the MIBK/IPA developer for 25 s while gently swirling the vial to maintain uniform development. Then transfer the sample immediately into pure IPA and continue swirling. The developer can be reused for all samples in the same batch, whereas fresh IPA should be used for each individual sample. Leave the sample in IPA for 60 s, then remove it and dry it carefully with a stream of nitrogen. During the full process, hold the sample securely with tweezers to avoid accidental dropping during waiting or transfer steps. Finally, inspect the developed structures under a light microscope to verify that the pattern is complete and free of obvious defects.

Shadow Evaporation

To begin shadow evaporation, place the sample holder on a hot plate at 100 °C for a few minutes in order to remove adsorbed water. This step improves the vacuum noticeably. Check the sample orientation carefully so that the shadow evaporation proceeds in the intended direction. For best results, load the sample into the load lock and pump it overnight, or longer if possible, before starting the deposition sequence.

Once the vacuum is satisfactory, transfer the sample holder into the main chamber for the first evaporation. Using electron-beam evaporation, deposit 60 nm of aluminum at an angle of 4°. Heat the aluminum gradually over approximately 5 – 10 min to avoid stressing the crucible and to promote stable evaporation. The crucible should contain a sufficient amount of aluminum; in practice, one to three pellets per evaporation are typically adequate.

After the first evaporation, transfer the sample back into the load lock and close the pumping line. Introduce pure oxygen until a pressure of 3.0(2) mbar is reached and oxidize the sample for 3 min. To terminate the oxidation, reopen the pumping line. In order to recover a vacuum on the order of 10^{-6} mbar afterward, it is important that the oxygen feed line itself has been evacuated thoroughly.

Once the load lock vacuum has recovered, transfer the sample back into the main chamber for the second evaporation. Deposit 100 nm of aluminum at an angle of 34° to complete the shadow-evaporation sequence.

A smooth aluminum surface is desirable for forming a well-defined Al_2O_3 layer. In practice, however, the deposited aluminum often showed a large number of dimples. Several measures were tested in an attempt to reduce them, but no decisive remedy was identified. There was some indication that higher evaporation rates, in particular values above 4 Å/s, may reduce the number of dimples, although the effect was not statistically convincing. Somewhat contrary to earlier expectations, a less extreme vacuum or a shorter time between the last vacuum break and the evaporation occasionally appeared to improve the surface. One possible interpretation is that a very clean sample mainly suppresses desorption of water or other contaminants during deposition, whereas a less ideal chamber condition may influence nucleation in a way that incidentally smooths the film. Since the observed differences remained small, the issue was not pursued further.

Lift-off is performed by placing the sample in a crimp-seal vial filled with pure acetone and heating it on a hot plate at about 60 °C for 1–3 h. Afterward, use a pipette to agitate the acetone carefully until all visible metal flakes have detached. Rinse the sample with IPA and dry it with nitrogen.

Plasma Etching

The penultimate fabrication step is plasma etching using the PlasmaPro 80 ICP RIE system by Oxford. Although this tool is not designed specifically for highly homogeneous etching, it can be operated in a regime that gives sufficiently uniform results by combining low table RF power with elevated pressure. An ignition step is used first to stabilize the plasma before switching to the less stable regime that favors homogeneous etching. The sample temperature is particularly important for obtaining the desired undercut. A table temperature of 100 °C proved to be a good compromise: it produces a sufficient undercut while keeping the etch slow enough to control the depth reliably. The process parameters that gave the best results are listed in Table 1.

A laser interferometer is used to monitor the etched depth. For this purpose, choose a spot on the sample without metal structures where the reflected intensity is as large as possible without reaching saturation.

During the etch, the signal follows an approximately cosine-like oscillation that can be used to estimate the removed thickness. The required number of periods m depends on the laser wavelength $\lambda = 632.8\,\mathrm{nm}$, the target depth $d = 500\,\mathrm{nm}$, and the refractive index $n = 1.81$ [42]. For the present process, the target value is

$$m = \frac{2 \cdot n \cdot d}{\lambda} = \frac{2 \cdot 1.81 \cdot 500\,\mathrm{nm}}{632.8\,\mathrm{nm}} = 2.86 \qquad (15)$$

periods. An example of the recorded laser intensity during etching is shown in Figure 19.

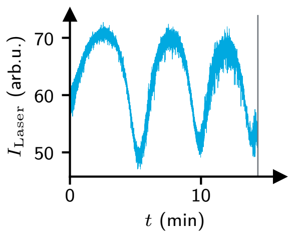


Figure 19 Recorded laser intensity $I_{\mathrm{Laser}}(t)$ during plasma etching. The process is terminated after approximately 2.8 periods, corresponding to the target etch depth.

Since the etch rate varies noticeably from sample to sample, each sample should ideally be etched individually.

Table 1 RIE recipe `MCBJ SSET v5 @ 100C`

Parameter	Stabilization	Ignition	Etching
Process time	5 min	5 s	30 min
Table heater	100 °C	-	-
Oxygen flow	50 sccm	-	-
Pressure	100 mTorr	ramp	250 mTorr
Table RF			
Power Demand	-	20(5) W	3(5) W
Max Reflected Power	-	5 W	2 W
Tolerance Time	-	7 s	-
Min DC Bias	-	1 V	-
AMU (C1, C2)	-	(41 %, 23 %)	-
ICP RF			
Power Demand	-	400(50) W	-
Max Reflected Power	-	100 W	-
Tolerance Time	-	10 s	-
AMU (C1, C2)	-	(64 %, 39 %)	-

Electrical contacting

Electrical contacting is one of the most delicate steps in the entire fabrication sequence. Many otherwise successful samples are lost at this stage, so patience and a calm working setup are more important than speed. In practice, it is usually better to contact fewer samples carefully than to rush several samples in one session.

Good working conditions matter. Stable hand positioning, appropriate lighting, and a comfortable bench height all make the process significantly easier. A setup that allows the operator to sit close to the sample while keeping both hands supported is particularly helpful. It is also worth minimizing avoidable sources of hand tremor before starting.

To fix the sample, adhesive strips[33] are used on a copper plate. This arrangement keeps the sample in place while still allowing some freedom to reposition the plate. Prepare 90 µm insulated copper wires soldered to post connectors in advance. The wire ends should be stripped, and the bias-line wires may be twisted. The wires should be kept as short as practical; typically, 4 – 5 cm is sufficient. The rear part of each post connector should be insulated with a varnish[34].

[33] e.g. tesafilm® standard by Tesa

[34] e.g. GE Low Temperature Varnish (59-C5-101) by Oxford Instruments

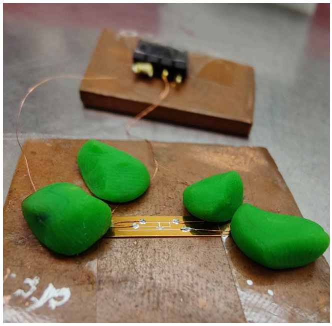


Figure 20 Example of a contacted sample, before adding strain relief.

Position the wire ends on the sample pads with the help of play dough. Then, using a sharpened toothpick, apply a small amount of silver paste to connect each wire to its pad. The viscosity of the silver paste is critical: if it is too fluid, shorts can form; if it is too viscous, it tends to remain on the toothpick instead of wetting the pad properly. In practice, using paste collected from the lid after shaking the bottle works well. Only a very small amount should be transferred, just enough to wet the pad. Afterward, apply two-component epoxy as strain relief for the wires. Here as well, the viscosity is important. If the epoxy is used too soon after mixing, it continues to flow and may spread into unwanted areas. Since the interaction between epoxy and silver paste is not fully understood, the two materials should be kept separate as much as possible. A useful approach is to test the epoxy on small trial dots and wait until it reaches a suitable consistency before applying it to the sample. The two sides of the sample are best glued separately, since the useful working window is fairly short.

After the sample has been mounted in the cryostat, the final step is to scratch the shorting bridge with an engraving tool. During this step, and also while closing the radiation shields, proper grounding is essential. Wearing a grounding bracelet is recommended. In dry conditions, additional precautions such as increasing the ambient humidity can further reduce the risk of electrostatic discharge.

Summary of key process parameters

Table 2 Key process parameters used for sample preparation.

Sacrificial layer	Durimide 115A
Spinning	spread cycle: 300 rpm for 30 s
	spin cycle: 5000 rpm for 90 s
	ramps for 3 s, `Program 3`
Soft baking	135 °C for 5 min in convection oven
Hard baking	400 °C for 30 min in vacuum oven
Spacing layer	MMA(8.5)MAA EL11
Spinning	spread cycle: 500 rpm for 5 s
	spin cycle: 2500 rpm for 90 s
	ramps for 3 s, `Program 1`
Soft baking	150 °C for 1 min on hot plate
Electron resist	950 PMMA A4
Spinning	spread cycle: 500 rpm for 5 s
	spin cycle: 5000 rpm for 60 s
	ramps for 3 s, `Program 2`
Hard baking	170 °C for 30 min in convection oven
Exposure	Crossbeam by Zeiss, Elphy Plus by Raith
Settings	10 kV acceleration, 5 mm working distance,
Aperture	30 µm and 120 µm aperture, high current,
	155 $\mu C/cm^2$ area dose
Development	25 s in 1:3 MIBK:IPA
	60 s in IPA
Shadow Evaporation	First layer: 60 nm Al at 4°
	Oxidation: 3.0 mbar O_2 for 3 min
	Second layer: 100 nm Al at 34°
Lift-off	1 h in acetone at 60 °C
Reactive Ion Etching	PlasmaPro 80 ICP RIE by Oxford
	`MCBJ SSET v5 @ 100C`, see Table 1
Electrical contacting	90 µm insulated copper wire, silver paste as
	electrical contact and epoxy as strain relief

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